

ABES ENGINEERING COLLEGE, GHAZIABAD

**DEPARTMENT OF ELECTRONICS & COMMUNICATION
ENGINEERING**



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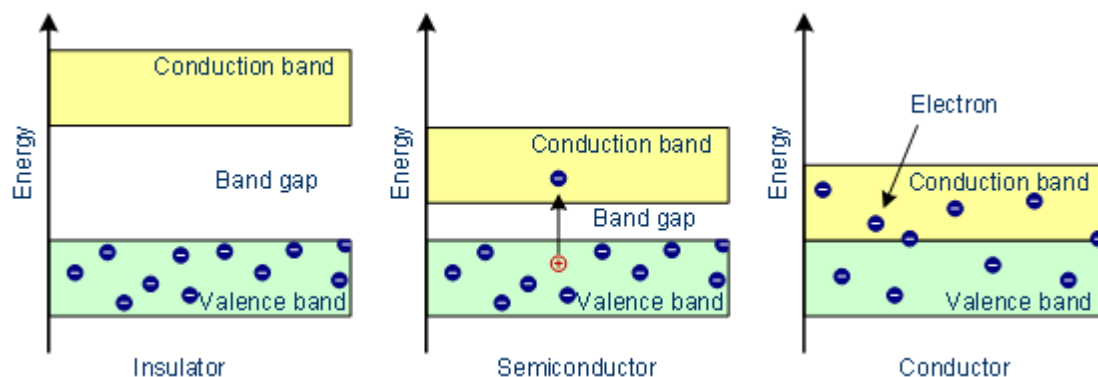
UNIT 1

DIODE AND ITS APPLICATIONS

1.P-N Junction Diode

1.1 Introduction

Valence Band- The electrons in the outermost shell are called valence electrons, Consequently, The energy band associated with the outermost shell is known as the **valence band**. The valence electrons are loosely bonded with the nucleus: therefore, a small amount of external energy is sufficient to free them from their atom. **Conduction Band-** When the valence electrons are freed from the atom, these free electrons are responsible for the conduction of current in solids and are therefore called **conduction electrons / free electrons**. The bond occupied by these electrons is called the **conduction band**. Therefore, the loosely bound valence electrons jump from the valence band to the conduction band when sufficient external energy is supplied to the atoms. The gap between valence band and conduction band is called the **forbidden energy gap(E_g)** as shown in Fig. 1.



Example

$E_g > 5 \text{ eV}$
 $E_g \text{ for Glass} = 10 \text{ eV}$
 $E_g \text{ for Diamond} = 6 \text{ eV}$

$E_g \approx 1 \text{ eV}$
 $E_g \text{ for Germanium} = 0.67 \text{ eV}$
 $E_g \text{ for GaAs} = 1.42 \text{ eV}$
 $E_g \text{ for silicon} = 1.12 \text{ eV}$

Fig. 1

Semiconductors

➤ A **semiconductor** is a material which has electrical conductivity between that of a conductor such as copper and that of an insulator such as glass. The conductivity of a semiconductor material increases with increasing temperature. All the elements of IV group of periodic table are semiconductors (C, Si, Ge) etc. The energy band gap between valence band and conduction band is less than 3 eV. There are two types of semiconductor material

- **Intrinsic or Pure semiconductors.**
- **Extrinsic or Doped Semiconductor**

Intrinsic Semiconductors-An intrinsic semiconductor material is chemically very pure and possesses poor conductivity. It has equal numbers of negative carriers (electrons) and positive carriers (holes) as shown in Fig. 2.

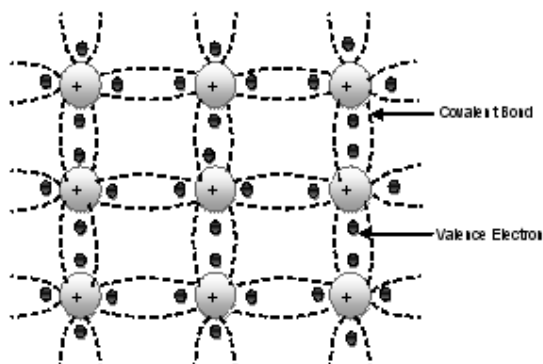


Fig 2.1 : Structure of intrinsic semiconductor at absolute zero temperature

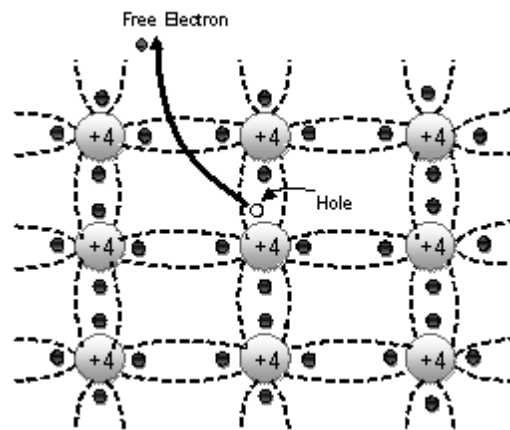
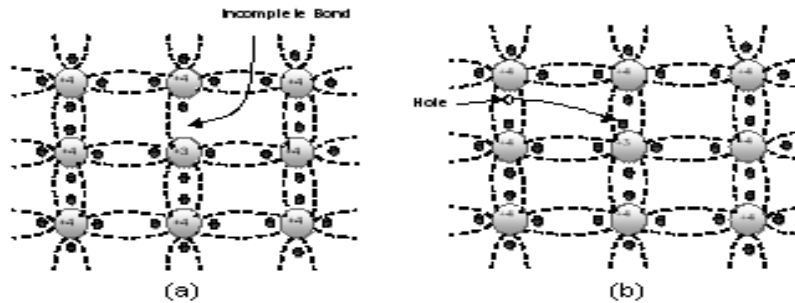


Fig 2.2: Generation of electron-hole pair in Intrinsic semiconductor

Fig. 2

Extrinsic Semiconductor- The conductivity of intrinsic semi-conductor can be increased by adding some suitable impurity atoms to the semiconductor. When a small amount of impurity atom is added to the intrinsic semiconductor, it is called extrinsic or impurity semi-conductor and the process of adding impurities to the semiconductor is known as **doping**.

P-Type- When a trivalent impurity atom is added to an intrinsic semiconductor, it is called P-type semiconductor. trivalent impurities are Boron, Aluminum, Gallium and Indium. They are known as acceptor impurities because the holes created can accept the electrons. It has been depicted in Fig. 3.



P-Type Semiconductor: - a) Boron added to Silicon b) Creation of a hole.

Fig. 3

N-Type- When a pentavalent impurity atom is added to an intrinsic semiconductor, it is called N-type semiconductor. Penta valent atoms are phosphorus, arsenic and antimony. They are called donor impurities because they donate one valence electron to the semiconductor crystal.

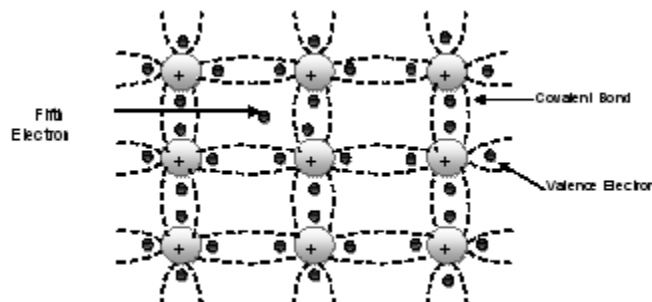


Fig. 4

Table 1 Different types of P-type and N-type material

	N-Type	P-Type
Doping	Pentavalent Atoms(P, As,Sb,Bi)	Trivalent Atoms (B,Al,Ga,In)
Majority Carriers	Electrons	Holes
Minority Carriers	Holes	Electrons
Ions	Positive / Donor Ions	Negative / Acceptor Ions

1.2 Working principle of PN Junction Diode

- Made by joining P-Type and N-Type Semiconductors
- Two terminal devices
- Unidirectional Device
- Non-Linear Device
- Basic Operation- as a **SWITCH** in electronic devices
- Applications: Rectifier, Zener Diode, Clipper, Clamper etc

Unbiased P-N Junction Diode- ($V = 0\text{ V}$) A P-N Junction formed by injecting P-Type and N-Type impurity atoms into the two halves of a single intrinsic semiconductor crystal is called **unbiased P-N-Junction**. The concentrations of the electrons and holes are different across the junction. This is known as **concentration gradient**; due to the concentration gradient across the junction, some changes start taking place as the P-N junction is formed. The unbiased PN junction diode has been shown in Fig. 5.

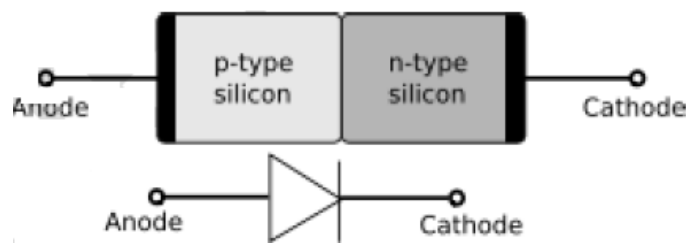


Fig. 5

- The holes from the P-region cross the junction & the N-region where there is a scarcity of holes. This is because the P-N junction tries to have an equal number of holes on both sides of the junction. This process is called **diffusion**.
- As soon as a hole from the P-region crosses the junction into the N-region, it confronts a free electron generated by a donor ion. This hole recombines with the free electron, and the free electron vanishes. The corresponding donor ion is now left without any electron associated with it. This donor ion is called an uncovered charge. In this manner a large number of uncovered positive charges are produced near the junction in the N-region.

- The electrons from the N-region diffuse into the P-region due to the concentration gradient across the junction. These electrons are recombined with the free holes associated with immobile acceptor ions near the junction, uncovered negative charges are produced in the P-region close to the junction.
- The uncovered negative charges create a negative electric field in the P-region & the uncovered positive charges create a positive electric field in the N-region. Now, if positively charged holes try to cross the junction into the N-region, the positive field repels them back into the P-region. Similarly the electrons are prevented from crossing the junction from N-region to the P-region, due to a repelling force exerted from the uncovered negative charges in the P-region.
- After some time, the transportation of mobile charges, electrons & holes stops due to a negative electric field in the P-region and a positive electric field in the N-region. The unbiased PN junction diode has been depicted in Fig. 6.

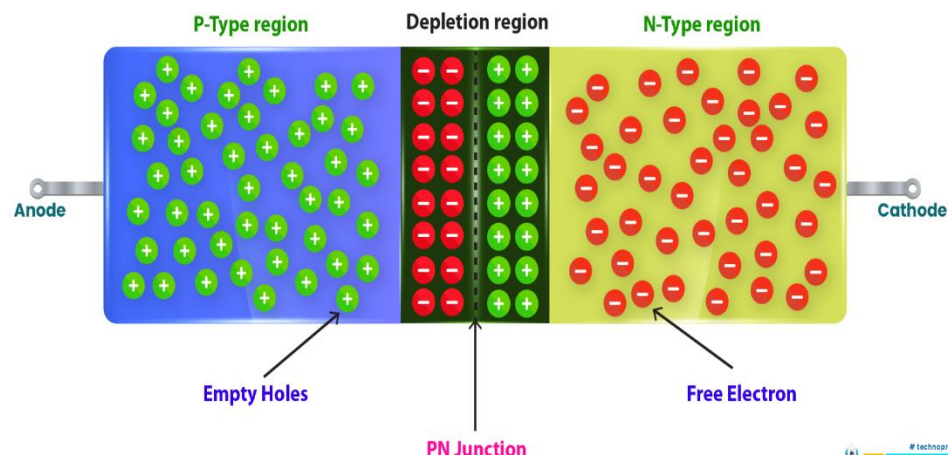


Fig. 6 Unbiased PN Junction

Depletion Region- The region of uncovered positive and negative ions is called the depletion region due to the “**depletion**” of free carriers in the region.“ In the absence of an applied bias across a semiconductor diode, the net flow of charge in one direction is zero.” V_B is the barrier voltage in PN junction diode.

$$V_B = 0.3V \text{ for Ge } V_B = 0.7V \text{ for Si}$$

Reverse Bias Condition ($V < 0V$) – When the negative terminal of the DC source is connected to the P-side & the positive terminal of the DC source is connected to N-side

of the P-N junction. The P-N junction is known as a **reverse biased junction**. The reverse bias PN junction diode is shown in Fig. 7.

1- The Current **does not flow because of majority carrier**.

2- Due to an **increase in positive & negative ions** around the junction, the electric field around the junction is increases, and as a result, the height of the potential barrier is **increased**.

3- The current that exist under reverse bias condition is called the reverse saturation current & is represented by I_s . (**Because of minority carriers only**)

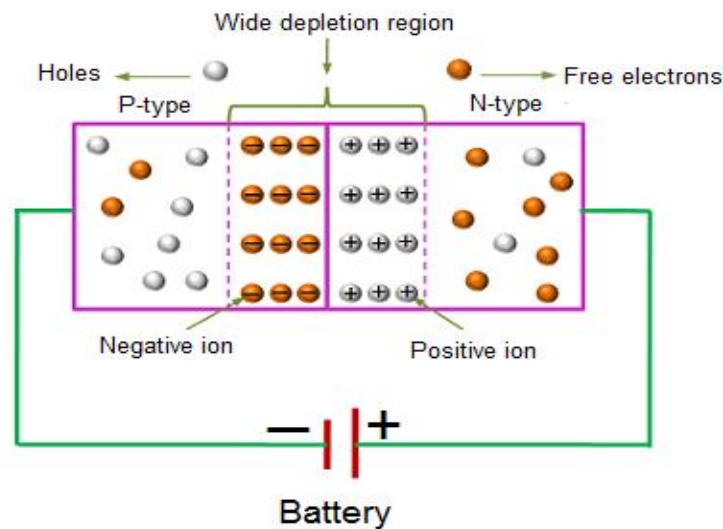


Fig. 7 Reverse Bias PN Junction

Forward Bias Junction ($V > 0V$)- If the positive terminal of the voltage source is connected to the P-side & the negative terminal of the source is connected to the N-side of the diode, then the diode is said to be biased in the **forward direction or forward biased**. The forward bias PN junction diode is shown in Fig. 8.

1- The current Flows because of **majority carriers only**.

2- **Depletion layer decreases with the increase of voltage because effect of external electric field is higher than internal potential barrier.**

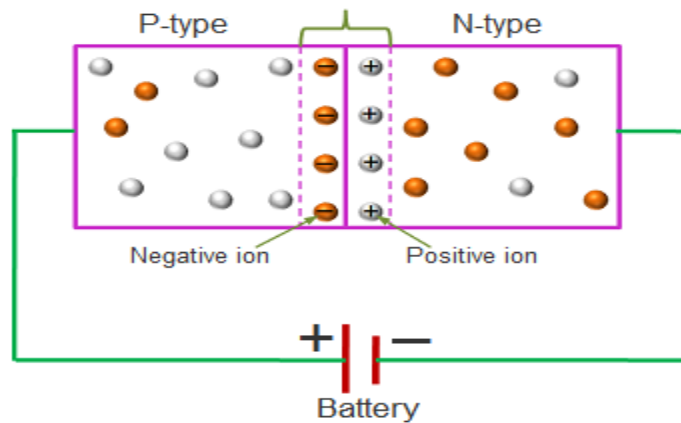


Fig. 8 Reverse Bias PN Junction

Table 2 Comparison between Forward and Reverse Bias PN Junction Diode

S No.	Forward Bias	Reverse Bias
1	P-Type → Positive Terminal N-Type → Negative Terminal	P-Type → Negative Terminal N-Type → Positive Terminal
2	Current Flows because of majority carriers.	Current Flows because of Minority carriers. → Reverse Saturation Current
3	Depletion Layer decreases with increasing voltage.	Depletion Layer increases with increasing voltage.
4	Less voltage required	High Voltage required

1.3 V-I Characteristics of P-N Junction Diode

V-I characteristics of P-N junction diode shows the graphical relationship between voltage and flowing current through diode

Forward Bias- When a forward voltage is applied at the terminals of a diode, the diode begins to conduct. During conduction, the cut in or threshold voltage exceeds the applied forward voltage. The threshold voltage for a germanium diode is 0.3V and for silicon diode is 0.7V. The forward current (milliampere range) initially increases linearly and then increases exponentially for high currents.

Reverse Bias- When a reverse voltage is applied, a reverse saturation current flows through the diode. The diode continues to be in the non conducting state until the reverse voltage drops below the Zener voltage. As the reverse voltage approximates the

peak inverse voltage a breakdown called as the '**Avalanche breakdown**' occurs. During the breakdown, the minority charge carriers ionize the stable atoms which are followed by a chain ionization to generate a large number of free charge carriers. Thus the diode becomes short circuited and gets damaged. The VI characteristics of PN junction diode has been shown in Fig. 9.

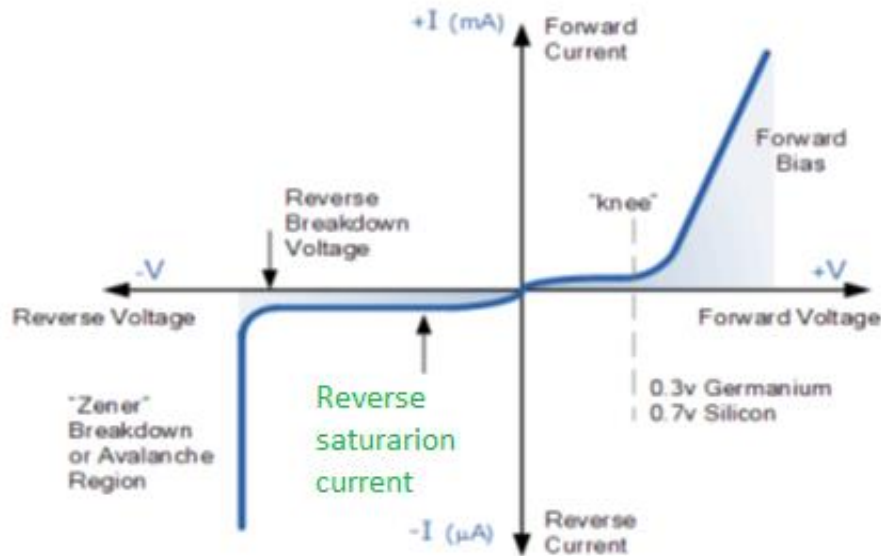


Fig. 9 VI Characteristics of PN Junction

Breakdown Mechanism

If the reverse-bias applied to a P-N junction is increased; a point will reach when the junction breaks down and reverse current rises sharply to a value limited only by the external resistance connected in series. This specific value of the reverse bias voltage breakdown voltage (V_Z). After breakdown voltage depends upon the width of depletion layer. The width of depletion layer depends upon the doping level. The following two processes cause junction breakdown due to the increase in reverse bias voltage. The breakdown mechanism is shown in Fig. 10.

1- Zener Breakdown-

- Occurred when a heavily doped junction is reverse biased.
- This is observed at $V < 6\text{ V}$.
- Field Ionization ($E = 3 \times 10^6\text{ V/cm}$) takes place in this mechanism.
- Tunneling of electrons [the valence electrons are pulled into conduction band]
- V-I characteristics with Zener breakdown is very sharp.
- Negative temperature coefficient. (When T increases V_Z decreases)

Tunneling Process-

Due to intense electric field ($E = 3 \times 10^6\text{ V/cm}$), the valence electrons are pulled into conduction band by breaking covalent bonds. These electrons become free electrons which are available for conduction. A large no. of such free electrons will constitute a large reverse current through the Zener diode and breakdown is said to have occurred due to the Zener effect. A current limiting resistance should be connected in series with the Zener diode to protect it against the damage due to excessive heating.

2- Avalanche Breakdown-

- Occurred when a lightly doped junction is reverse biased.
- Multiplication of electrons takes place in avalanche breakdown.
- This is observed at $V > 8\text{ V}$
- Impact Ionization.
- This V-I characteristics with the avalanche breakdown increases gradually.

- Positive temperature coefficient of voltage. (When T increases V_Z increases).

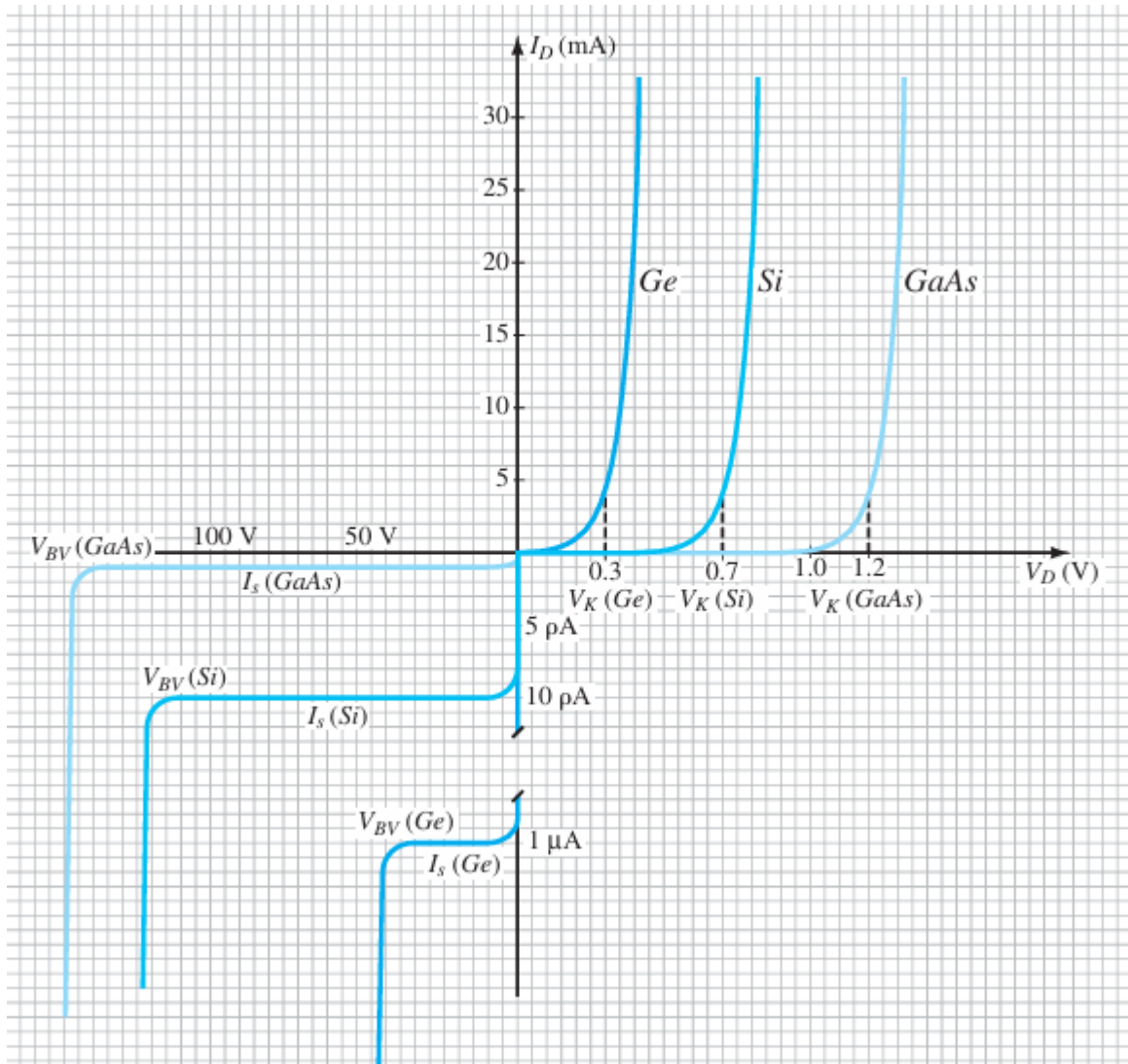


Fig. 10 Breakdown mechanism of PN Junction

1.3 Applications of PN Junction Diode

1.3.1 Rectifier

- A rectifier is an electrical component that converts alternating current (AC) to direct current (DC). A rectifier is analogous to a one-way valve that allows an electrical current to flow in only one direction.

1.3.1.1 Half wave Rectifier

A half-wave rectifier is an electronic circuit that converts alternating current (AC) to pulsating direct current (DC) by allowing only one half of the input AC waveform to pass. The circuit diagram for the half wave rectifier is shown in Fig. 11.

Circuit Diagram

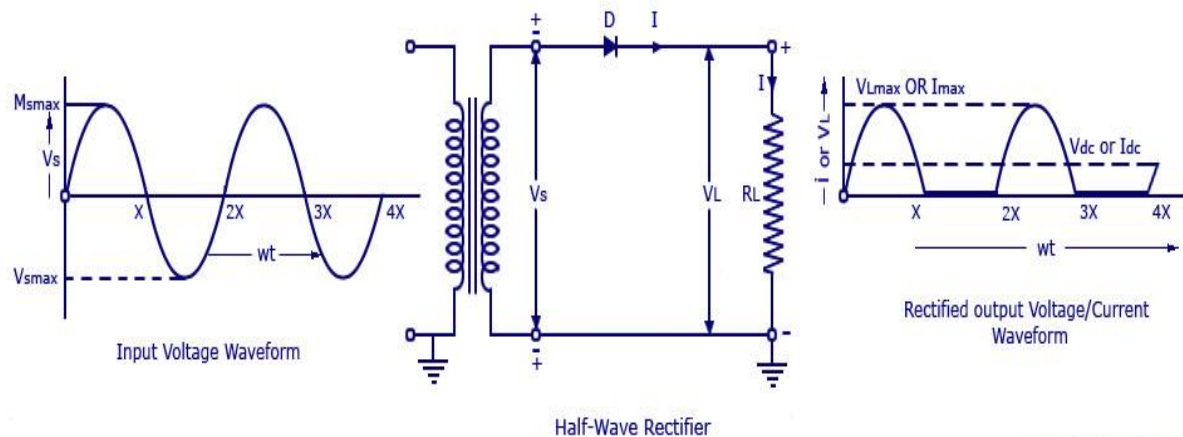


Fig. 11 Half Wave Rectifier

Working

The ac voltage across the secondary winding changes polarities after every half cycle of input wave. During the positive half-cycles of the input ac voltage *i.e.* when upper end of the secondary winding is positive w.r.t. its lower end, the diode is forward biased and therefore conducts current. If the forward resistance of the diode is assumed to be zero (in practice, however, a small resistance exists)

the input voltage during the positive half-cycles is directly applied to the load resistance R_L , making its upper end positive w.r.t. its lower end. The waveforms of the output current and output voltage are of the same shape as that of the input ac voltage. During the negative half cycles of the input ac voltage *i.e.* when the lower end of the secondary winding is positive w.r.t. its upper end, the diode is reverse biased and so does not conduct. Thus, during the negative half cycles of the input ac voltage, the current through and voltage across the load remains zero. The reverse current, being very small in magnitude, is neglected. Thus, for the negative half cycles no power is delivered to the load. Thus, the output voltage (V_L) developed across load resistance R_L is a series of positive half cycles of alternating voltage, with intervening very small constant negative voltage levels, It is obvious from the figure that the output is not a steady dc, but only a pulsating dc wave. To make the output wave smooth and useful in a DC power supply, we have to use a filter across the load. Since only half-cycles of the input wave are used, it is called a ***half wave rectifier***.

PERFORMANCE PARAMETERS OF HALF WAVE RECTIFIER

Vdc or Average Value:

The current wave of half wave rectifier is shown in Fig. 12.

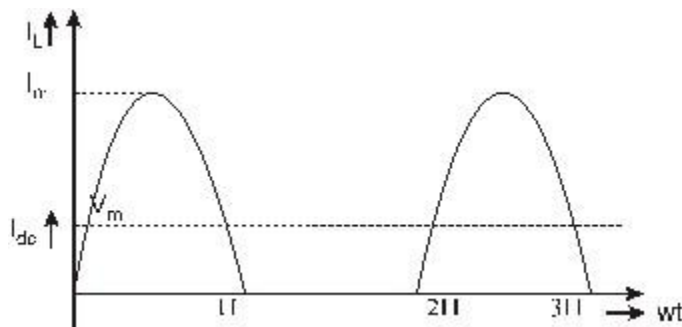


Fig. 12

$$I_L = \begin{cases} I_m \sin wt, & \text{for } 0 \leq wt \leq \pi \\ 0, & \text{for } \pi \leq wt \leq 2\pi \end{cases}$$

Where I_m is the peak value of load current.

THE DC VALUE:

$$I_{dc} = \frac{\text{Area}}{\text{Base}} = \frac{\int_0^{2\pi} i_L d(wt)}{2\pi}$$

$$= \frac{1}{2\pi} \int_0^{2\pi} i_L d(wt)$$

$$= \frac{1}{2\pi} \int_0^{\pi} I_m \sin w + d(wt) + \frac{1}{2\pi} \int_{\pi}^{2\pi} 0 d(wt)$$

$$= \frac{I_m}{2\pi} [-\cos wt]_0^{\pi} + 0 = \frac{I_m}{\pi}$$

$$I_{dc} = \frac{I_m}{\pi} \dots\dots\dots$$

The total rms value:

The RMS (Root Mean Square) value of a half-wave rectifier is the effective value of the output voltage or current after it has been rectified, which is the square root of the average of the squared output waveform over one full cycle.

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} i_m^2 d(\omega t)}$$

$$= \sqrt{\frac{1}{2\pi} \left[\int_0^{2\pi} I_m^2 \sin^2 \omega t d(\omega t) + \int_0^{2\pi} 0 d(\omega t) \right]}$$

$$= \sqrt{\frac{I_m^2}{2\pi} \int_0^{\pi} \frac{(1 - \cos 2\omega t)}{2} d(\omega t)} = \frac{I_m}{2}$$

RIPPLE FACTOR (r):

The ripple factor of a half-wave rectifier is a measure of the unwanted alternating current (AC) component remaining in the output.

$$i = i_L - I_{dc}$$

$$I'_{rms} = \sqrt{I_{rms}^2 - I_{dc}^2} \quad \text{where } I'_{rms} = \text{rms value of AC Component}$$

$$r = \frac{I'_{rms}}{I_{dc}} = \sqrt{\frac{(I_{rms}^2 - I_{dc}^2)}{I_{dc}^2}}$$

$$= \sqrt{\left(\frac{I_{rms}}{I_{dc}}\right)^2 - 1} = \sqrt{\left(\frac{\frac{I_m}{2}}{\frac{I_m}{\pi}}\right)^2 - 1} = 1.21$$

$$r = 1.21$$

PIV

The maximum voltage across a reverse bias diode is known as Peak Inverse Voltage (PIV). For Half Wave Rectifier PIV is V_m .

RECTIFICATION EFFICIENCY (η):

The efficiency of a rectifier is defined as the ratio of dc output power to input power.

$$\begin{aligned}\eta &= \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_L + r_f)} \\&= \left(\frac{I_{dc}}{\frac{I_m}{2}} \right)^2 \frac{R_L}{R_L + r_f} \\&= \left(\frac{\frac{I_m}{\pi}}{\frac{I_m}{2}} \right)^2 \frac{R_L}{R_L + r_f} \\&= 0.406 \frac{R_L}{R_L + r_f} = \frac{40.6}{1 + \frac{r_f}{R_L}} \%\end{aligned}$$

$$\eta\% = \frac{40.6}{1 + \frac{r_f}{R_L}} \%$$

Where r_f is the small forward resistance of diode.

In ideal diode $r_f = 0$

$$\eta\% = 40.6\% \dots\dots\dots$$

1.3.1.2 Full wave Rectifier

a) Centre Tapped Rectifier

A second popular full-wave rectifier appears in the Fig. 13. With only two diodes but requiring a center-tapped (CT) transformer to establish the input signal across each section of the secondary of the transformer. During the positive

portion of v_i applied to the primary of the transformer, the network will appear as shown in following figure. D_1 assumes the short-circuit equivalent and D_2 the open-circuit equivalent, as determined by the secondary voltages and the resulting current directions. The output voltage appears as shown in Fig 14.

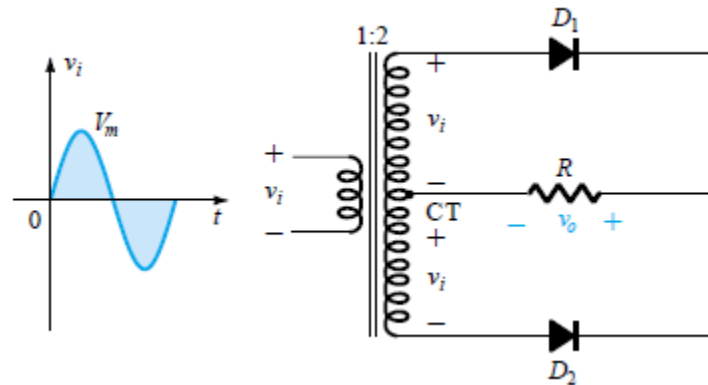


Fig. 13

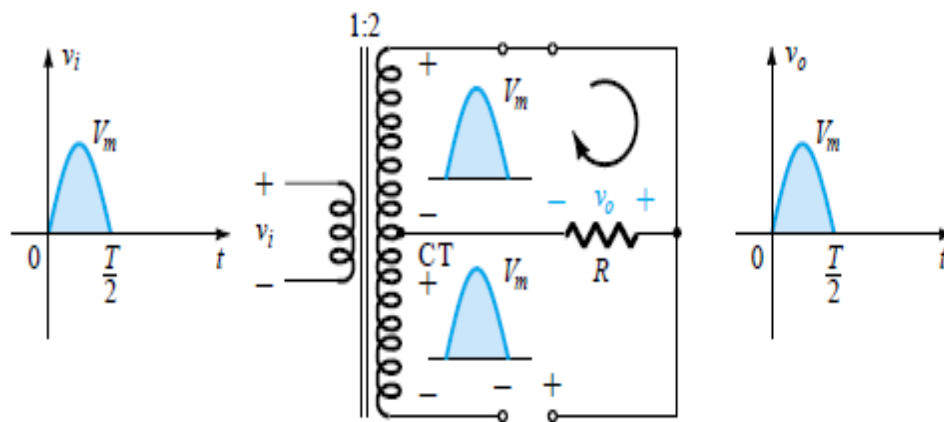


Fig. 14

During the negative portion of the input the network appears as shown in Fig. 15, reversing the roles of the diodes but maintaining the same polarity for the voltage across the load resistor R . The net effect is the same output as that appearing in Fig 16. with the same dc levels.

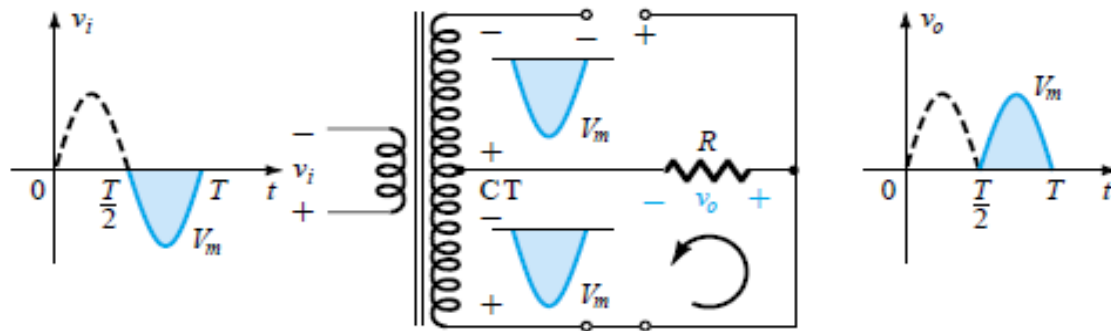


Fig. 15

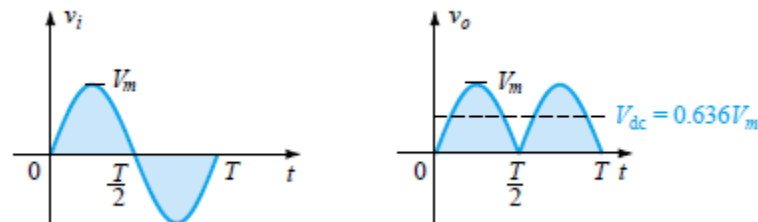


Fig. 16

PERFORMANCE PARAMETERS OF FULL WAVE RECTIFIER

PEAK INVERSE VOLTAGE (PIV)

The maximum voltage across a reverse bias diode is known as Peak Inverse Voltage (PIV).

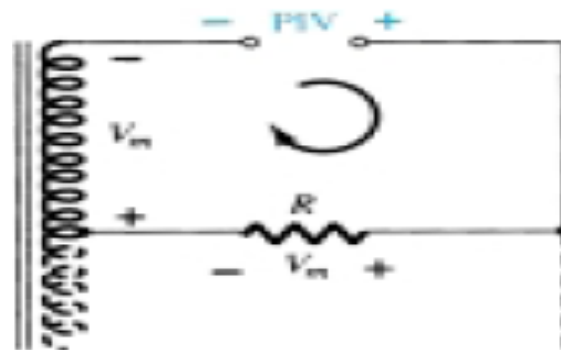


Fig. 17

The network will help us determine the net PIV for each diode for this full-wave rectifier. Inserting the maximum voltage for the secondary voltage and V_m as

$$\begin{aligned} \text{PIV} &= V_{\text{secondary}} + V_R \\ &= V_m + V_m \end{aligned}$$

established by the adjoining loop will result in

$$\text{PIV} \geq 2V_m$$

Ripple Factor

The ripple factor of a half-wave rectifier is a measure of the unwanted alternating current (AC) component remaining in the output.

$$\text{Ripplefactor} = \frac{I_{ac}}{I_{dc}} \quad I_{rms} = \sqrt{I_{dc}^2 + I_{ac}^2} \quad \frac{I_{ac}}{I_{dc}} = \sqrt{\frac{I_{rms}^2}{I_{dc}^2} - 1}$$

For full wave rectifier: $I_{rms} = \frac{I_m}{\sqrt{2}} \quad I_{dc} = \frac{2I_m}{\pi} \quad \text{Ripple factor} = 0.48$

b) Bridge Rectifier

A bridge rectifier makes use of four diodes in a bridge arrangement to achieve full-wave rectification. This is a widely used configuration, both with individual diodes wired as shown and with single component bridges where the diode bridge is wired internally. The circuit diagram for Bridge rectifier is shown in Fig. 18.

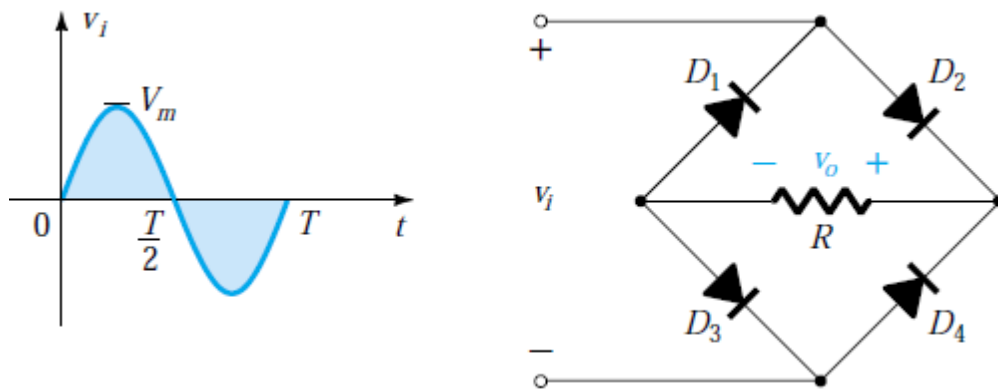


Fig. 18

Current Flow in the Bridge Rectifier

Positive half cycle

During the Positive half cycle D_2 and D_3 are conducting while D_1 and D_4 are in the “off” state. The net result is the configuration of Fig., with its indicated current and polarity across R . Since the diodes are ideal the load voltage is $v_o = v_i$, as shown in the Fig. 19.

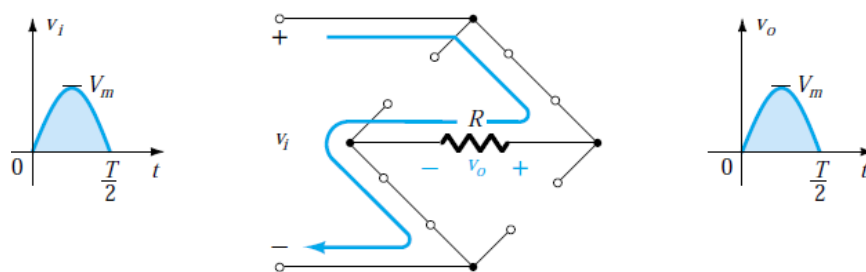


Fig. 19

Conduction path for the positive region of v_i .

Negative half cycle

During the negative half cycle D_1 and D_4 are conducting while D_2 and D_3 are in the “off” state. The net result is the configuration of, with its indicated current and polarity across R . Since the diodes

are ideal the load voltage is $v_o = v_i$, as shown in Fig. 20.

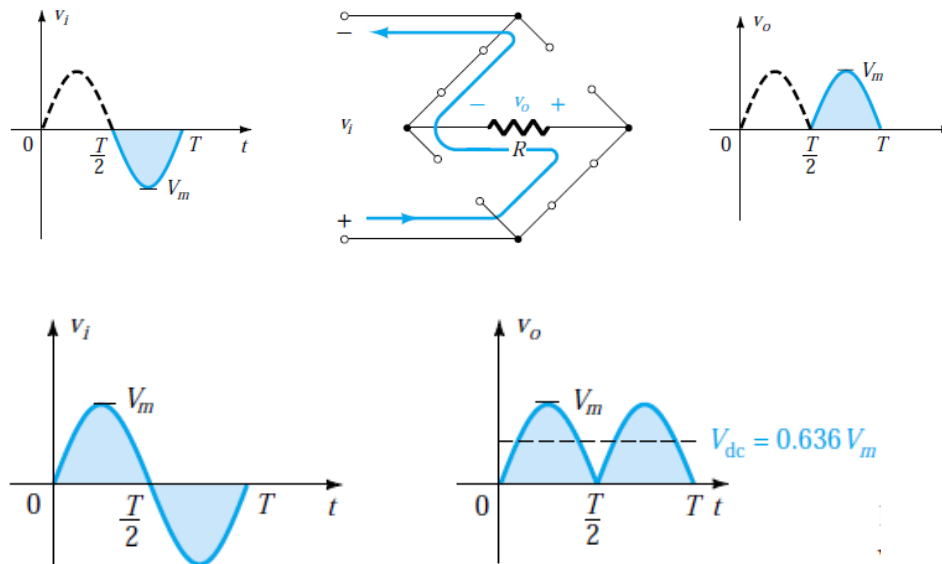


Fig. 20

PEAK INVERSE VOLTAGE (PIV)

The required PIV of each diode (ideal) can be determined from Fig. 21. Obtained at the peak of the positive region of the input signal. For the indicated loop the maximum voltage across R is V_m and the PIV rating is defined by

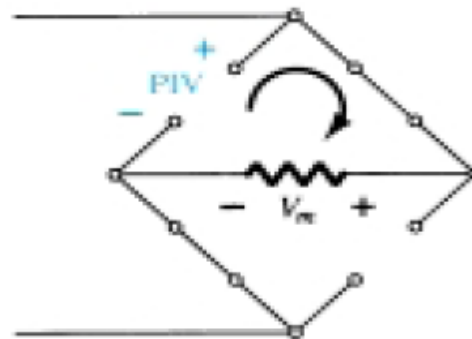


Fig. 21

1.3.2 Clippers and Clampers

Clipper

A circuit which cutoff voltage above or below are both specified level is called clipper. A clipper which removes a portion of positive half cycle of the input signal is called positive clipper. A clipper circuit that removes the negative half cycle is called negative clipper.

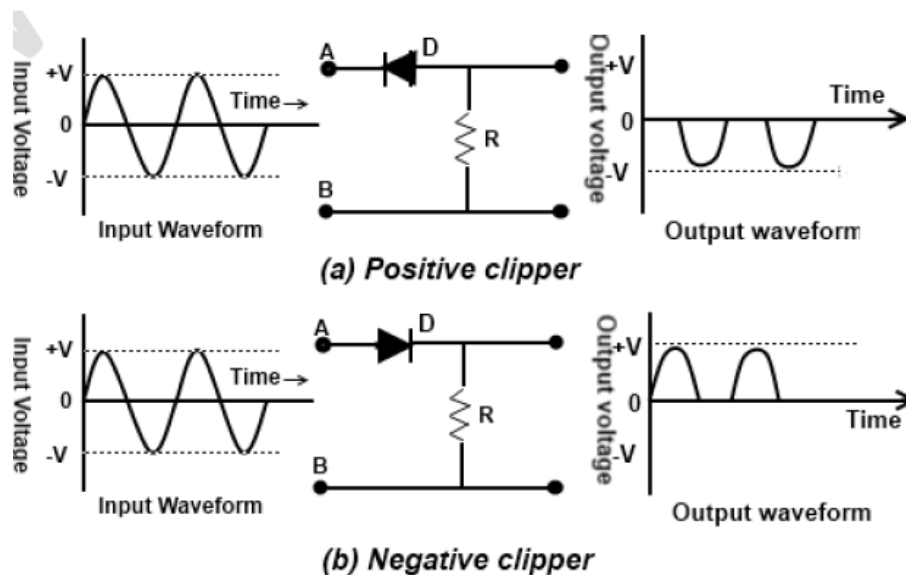


Fig. 22

Figure 22 (a) shows the circuit of a positive clipper. It consists of a diode D and a resistor R with output taken across the resistor. During positive half cycle the input voltage, the terminal A is positive with respect to B . This reverse biases the diode and it acts as an open switch. Therefore, all the applied voltage drops across the diode and none across the resistor. As a result of this, there is no output voltage during the positive half cycle of the input voltage. During the negative half cycle of the input voltage, the terminal B is positive with respect to A . Therefore, it forward biases the diode and it acts as a closed switch. Thus, there is no voltage drop across diode. During negative half cycle of the input voltage. **Figure 22 (b)** shows the waveform of the input voltage. During the

positive half cycle of the voltage, the terminal A is positive with respect to the terminal B. Therefore, the diode is forward biased; as a result, all the input voltage appears across the resistor. During negative half cycle of the input voltage, the terminal B is positive with respect to the terminal A. Therefore, the diode is reverse biased and hence there is no voltage drop across the resistor during negative half cycle.

Q 1. Determine v_o for the network of given Figure 23.

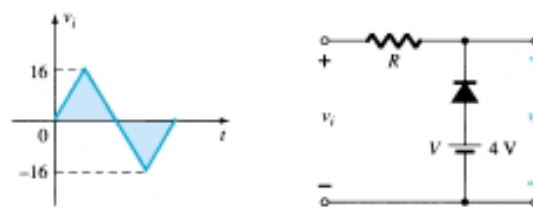


Fig. 23

Answer The polarity of the dc supply and the direction of the diode strongly suggest that the diode will be in the “on” state for the negative region of the input signal.

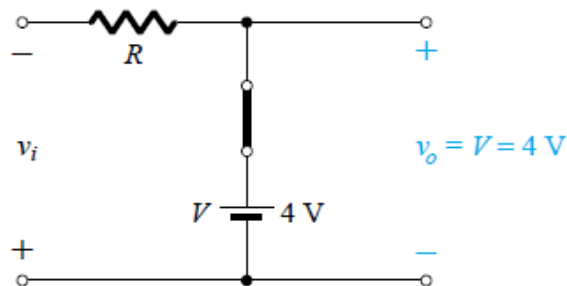


Fig:24 V_o for the negative region

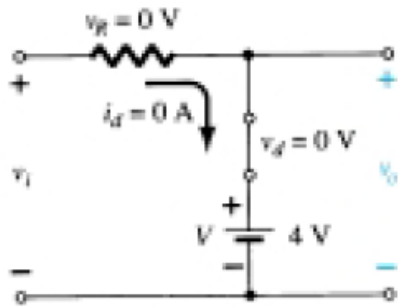


Fig: 25 Determining the transition level

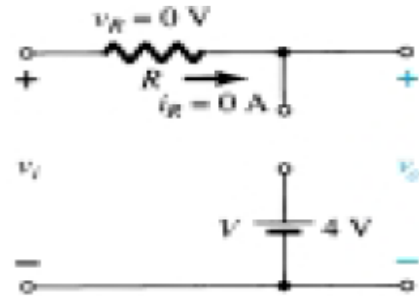


Fig:26 Determine V_o .

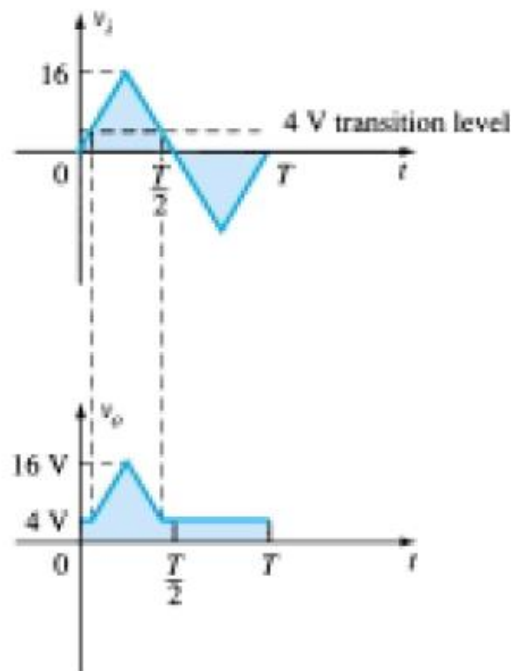


Fig:27 input and Output Waveforms

Q.2 Determine the output waveform for the given network in Fig. 28.

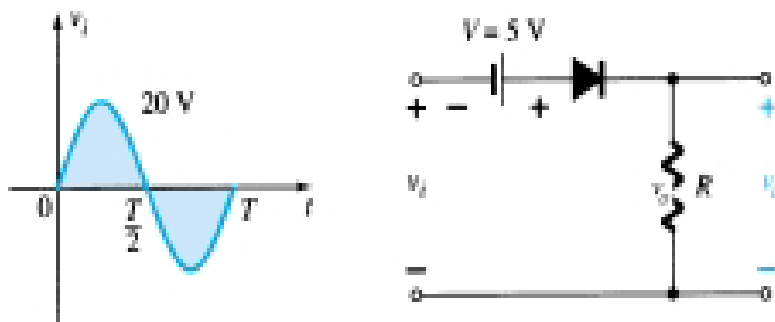


Fig:28

Answer: The diode will be in the “on” state for the positive region of v_i —especially when we note the aiding effect of $V=5\text{ V}$.

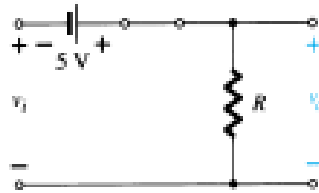


Fig. 29 V_o with the diode in ON state

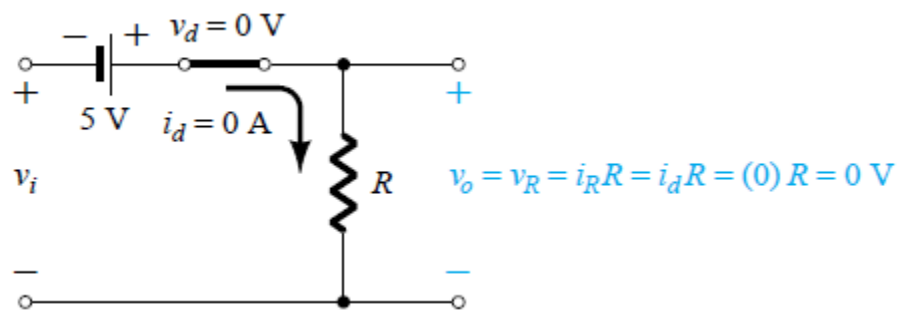


Fig 30: Determining the transition level

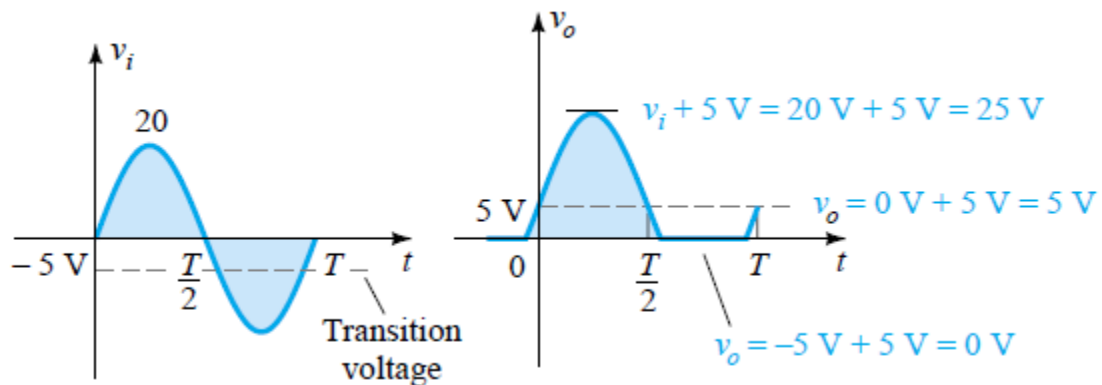
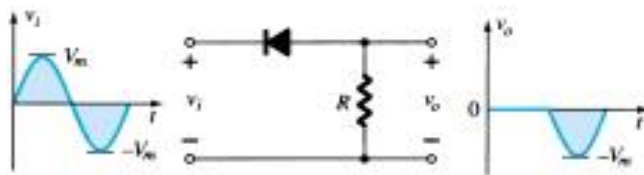


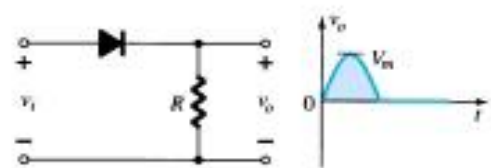
Fig 31: Input and Output Waveforms

Simple Series Clippers (Ideal Diodes)

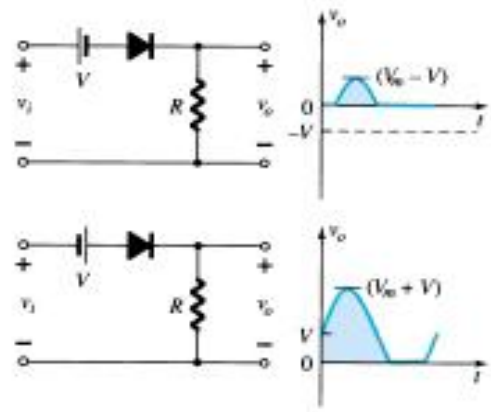
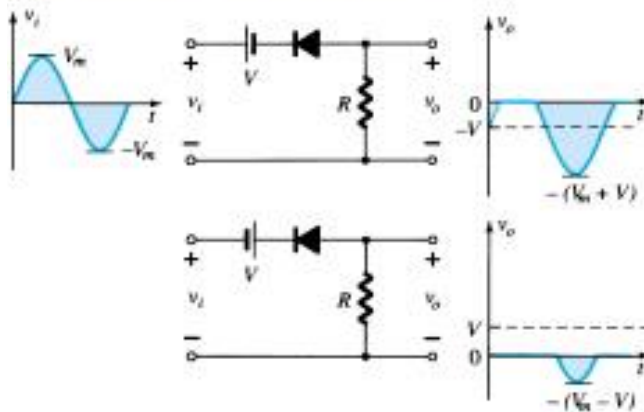
POSITIVE



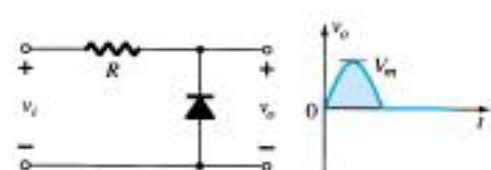
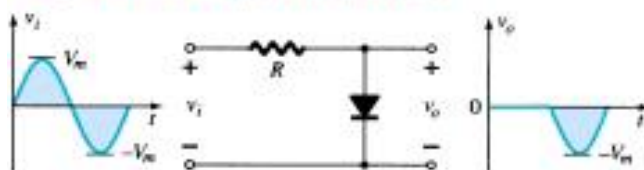
NEGATIVE



Biased Series Clippers (Ideal Diodes)



Simple Parallel Clippers (Ideal Diodes)



Biased Parallel Clippers (Ideal Diodes)

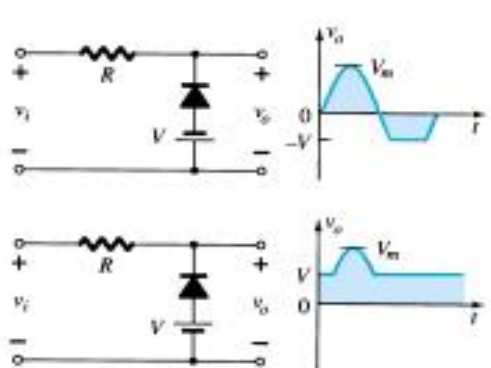
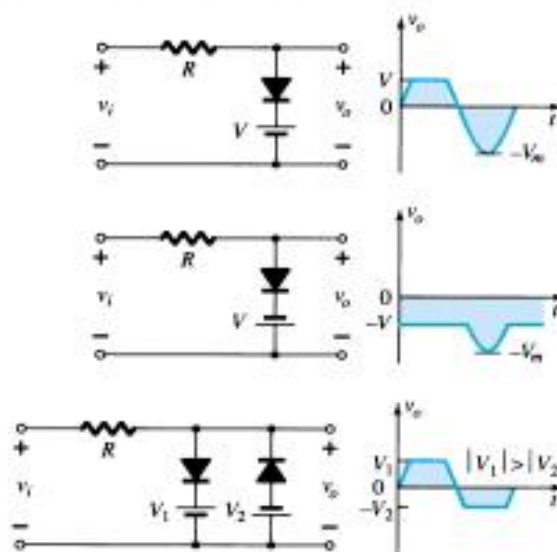


Fig:32

Clamper

- The *clamping* network is one that will “clamp” a signal to a different dc level. The network must have a capacitor, a diode, and a resistive element, but it can also employ an independent dc supply to introduce an additional shift. The magnitude of R and C must be chosen such that the time constant $\zeta=RC$ is large enough to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is non conducting. Throughout the analysis we will assume that for all practical purposes the capacitor will fully charge or discharge in five time constants.
- The network of Fig. 33 will clamp the input signal to the zero level (for ideal diodes). The resistor R can be the load resistor or a parallel combination of the load resistor and a resistor designed to provide the desired level of R .

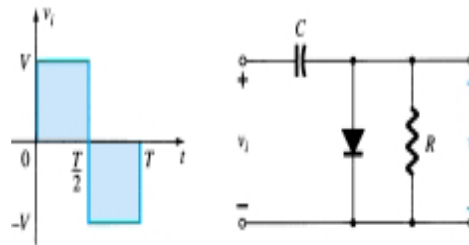


Fig:33

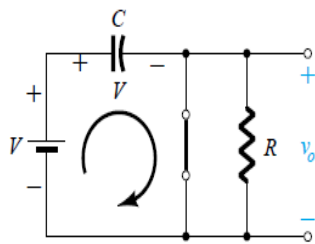


Fig: 34

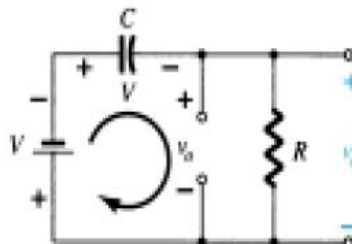


Fig: 35

- During the interval $0 < t < T/2$ the network will appear as shown in Fig. 34 With the diode in the “on” state effectively “shorting out” the effect of the resistor R . When the input switches to the $-V$ state the network will appear with

the open-circuit equivalent for the diode determined by the applied signal. Since v_o is in parallel with the diode and resistor, it can also be drawn in the alternative position shown in Fig. 35. Applying Kirchhoff's voltage law around the input loop will result in

$$-V -V -v_o = 0$$

$$v_o = -2V$$

- The negative sign resulting from the fact that the polarity of $2V$ is opposite to the polarity defined for v_o . The resulting output waveform appears in Fig. 36 with the input signal. The output signal is clamped to 0 V for the interval 0 to $T/2$ but maintains the same total swing ($2V$) as the input.
- For a clamping network:

In general, the following steps may be helpful when analyzing clamping networks:

- 1. Start the analysis of clamping networks by considering that part of the input signal that will forward bias the diode.**
- 2. During the period that the diode is in the "on" state, assume that the capacitor will charge up instantaneously to a voltage level determined by the network.**
- 3. Assume that during the period when the diode is in the "off" state the capacitor will hold on to its established voltage level.**
- 4. Throughout the analysis maintain a continual awareness of the location and reference polarity for v_o to ensure that the proper levels for v_o are obtained.**
- 5. Keep in mind the general rule that the total swing of the total output must match the swing of the input signal.**

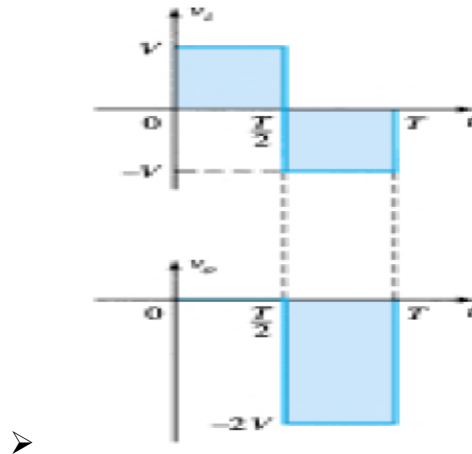


Fig: 36

Q1. Determine v_o for the network of Fig. 37 for the input indicated

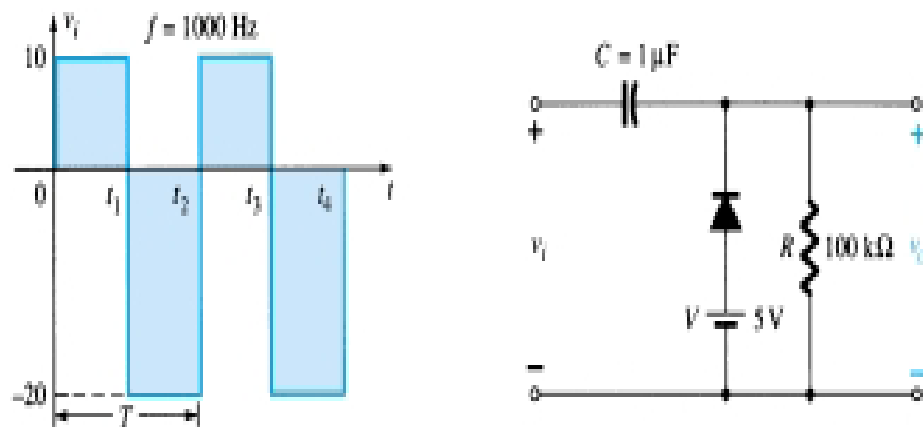


Fig: 37

Answer- The output is across R , but it is also directly across the 5-V battery. The circuit diagram is shown in Fig. 38.

$$-20 \text{ V} - VC - 5 \text{ V} = 0$$

$$VC = 25 \text{ V}$$

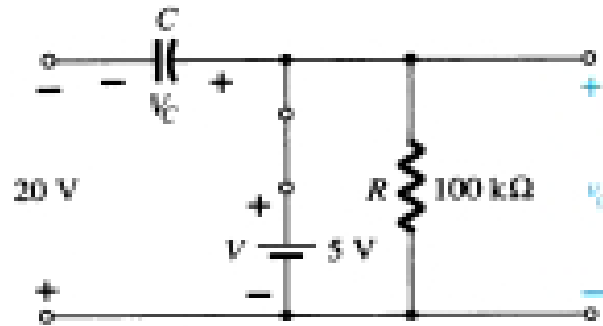


Fig: 38

The open-circuit equivalent for the diode will remove the 5-V battery from having any effect on v_o , and applying Kirchhoff's voltage law around the outside loop of the network will result in shown in Fig. 39.

$$-10 \text{ V} - 25 \text{ V} - v_o = 0$$

$$v_o = 35 \text{ V}$$

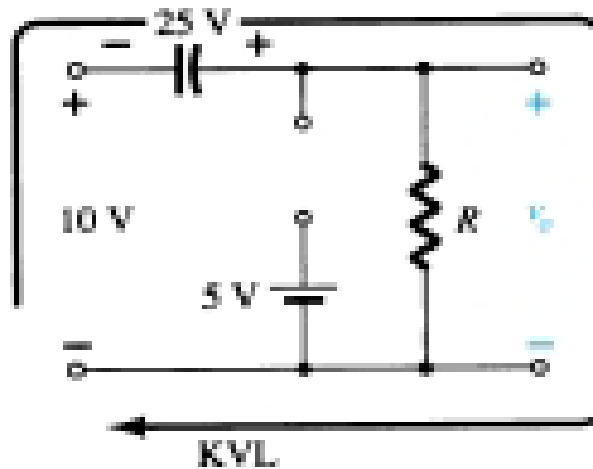


Fig: 39

The output waveforms are shown in Fig.40.

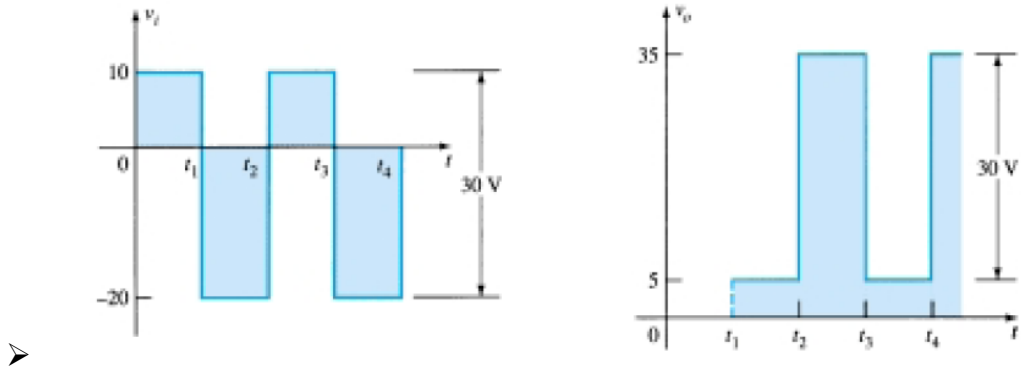
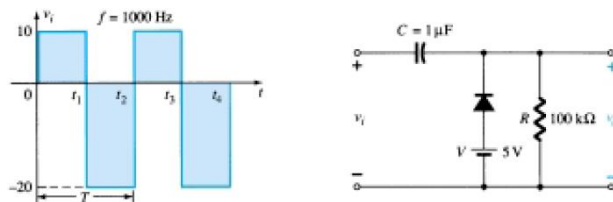


Fig: 40

Q2. Determine v_o for the given network using a silicon diode with $V_T=0.7V$



Answer V_o can be determined by Kirchhoff's voltage law in the output section

$$-5 \text{ V} - 0.7 \text{ V} - v_o = 0$$

$$v_o = 5 \text{ V} - 0.7 \text{ V} = 4.3 \text{ V}$$

For the input section Kirchhoff's voltage law will result in

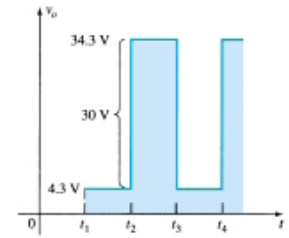
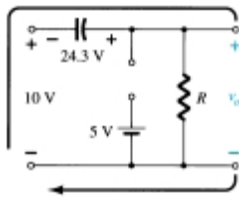
$$-20 \text{ V} - V_C - 0.7 \text{ V} - 5 \text{ V} = 0$$

$$V_C = 25 \text{ V} - 0.7 \text{ V} = 24.3 \text{ V}$$

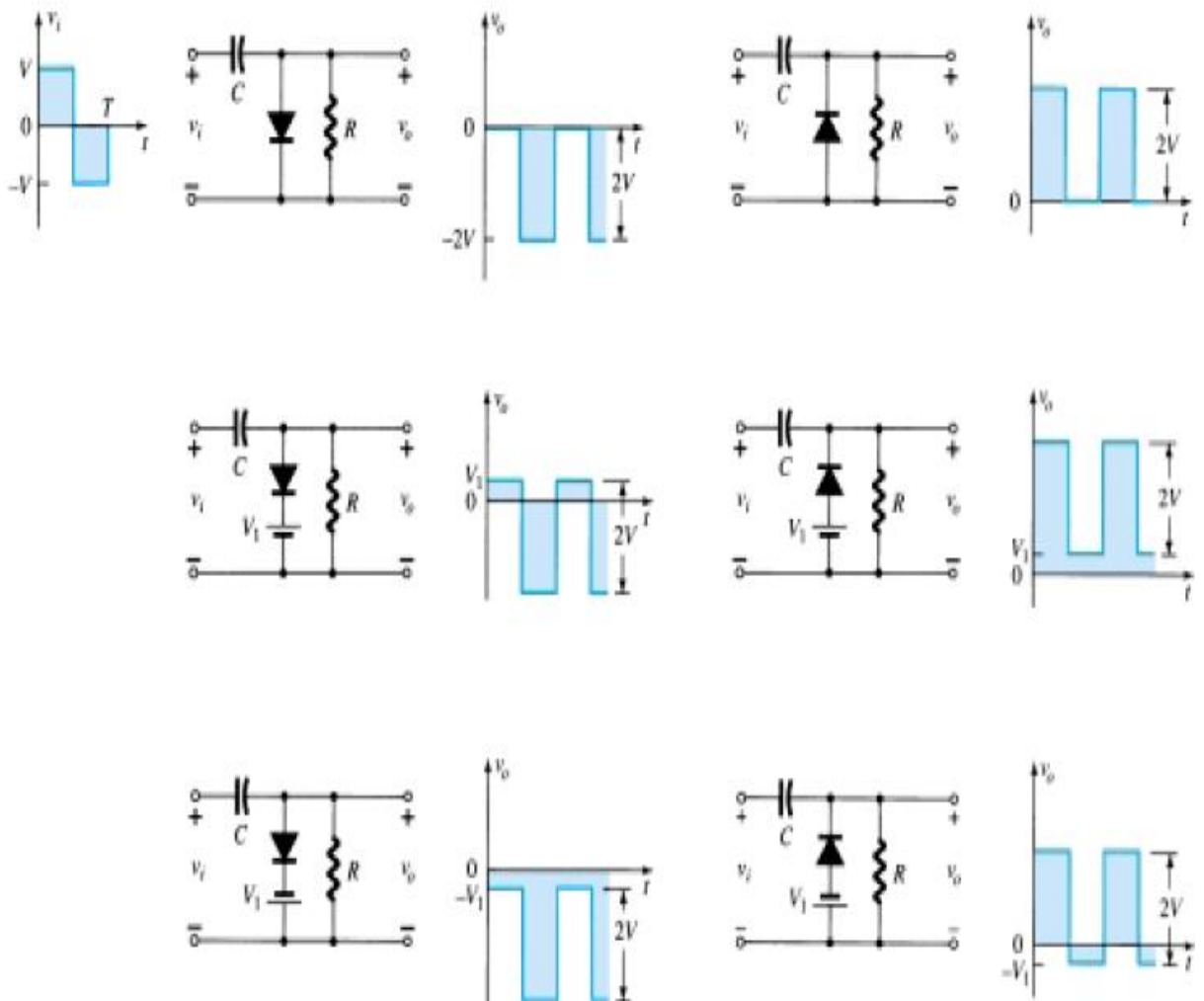
Applying Kirchhoff's voltage law yields

$$-10 \text{ V} - 24.3 \text{ V} - v_o = 0$$

$$v_o = 34.3 \text{ V}$$



A number of clamping circuits and their effect on the input signal are shown below



2.Special Purpose Devices

2.1 Zener Diode

Zener Diode

Zener diode is a reverse biased heavily doped P-N junction diode which operates in the breakdown region. The reverse breakdown of a P-N junction may occur either due to zener effect or avalanche effect. **Conduction direction of zener diode** - For the zener diode the direction of conduction is opposite to that of the arrow in the symbol. Zener diode equivalent circuit symbol and equivalent circuit is shown in Fig. 41.

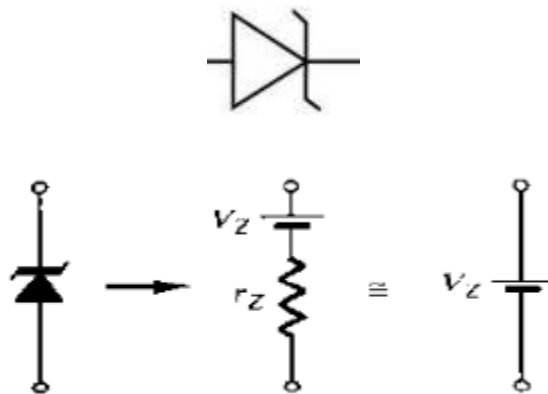


Fig: 41

The complete equivalent circuit of the zener diode in the zener region includes a small dynamic resistance and dc battery equal to the zener potential. Generally, the zener resistance value is quite small than external resistance of the circuit in which zener diode is connected. Therefore, zener resistance may be neglected generally.

On State ($V > V_Z$) If the reverse bias voltage across a zener diode is equal to or more than breakdown voltage V_Z , the current will increase sharply. Hence, in this region the current will be almost vertical.

Off State ($0 < V < V_Z$) If the reverse voltage across the zener diode is less than V_Z but greater than 0V, the zener diode will be in the off state.

Zener diode as voltage regulator

The analysis of networks employing Zener diodes is quite similar to that applied to the analysis of semiconductor diodes in previous sections. First the state of the diode must be determined followed by a substitution of the appropriate model and a determination of the other unknown quantities of the network. Unless otherwise specified, the Zener model to be employed for the "on" state will be as shown in Fig. 42. For the "off" state as defined by a voltage less than V_Z .

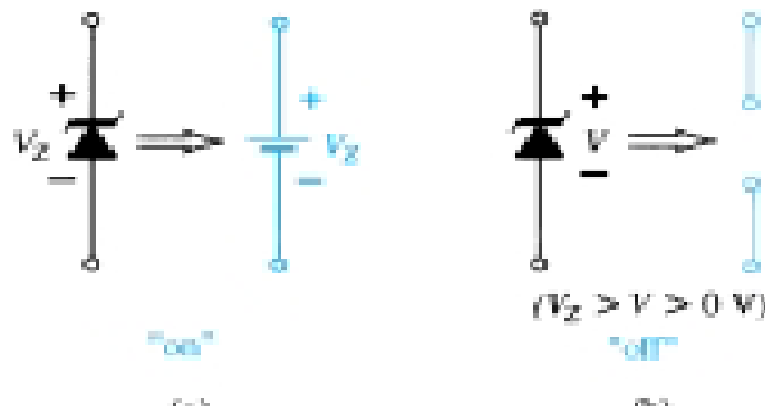


Fig: 42

The analysis of Zener diode as a voltage regulator is considered using the basic circuit shown in Fig. 43. With response of the Zener diode to different operating conditions. The analysis is first for fixed quantities followed by a fixed supply voltage and a variable load and finally a fixed load with a variable supply.

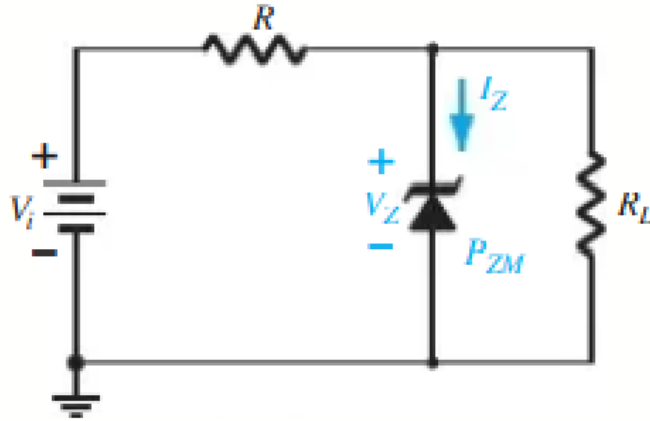


Fig: 43

V_i and R Fixed

The simplest of Zener diode regulator networks appears in Fig. 44. The applied dc voltage is fixed, as is the load resistor. The analysis can fundamentally be broken down into two steps.

1. Determine the state of the Zener diode by removing it from the network and calculating the voltage across the resulting open circuit. Applying step 1 to the network of Fig. 43 results in the network of Fig. 44, where an application of the voltage divider rule results in eq. (1).

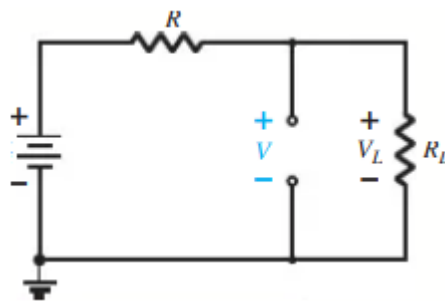


Fig: 44

$$V = V_L = \frac{R_L V_i}{R + R_L}$$

(1)

If $V > V_Z$, the Zener diode is on, and the appropriate equivalent model can be substituted. If $V < V_Z$, the diode is off, and the open-circuit equivalence is substituted.

2. Substitute the appropriate equivalent circuit and solve for the desired unknowns. For the network of Fig. 43, the "on" state will result in the equivalent network of Fig. 45.

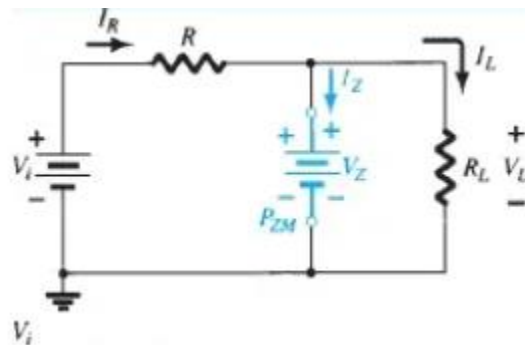


Fig: 45

The Zener diode current must be determined by an application of Kirchhoff's current law. That is,

$$I_R = I_Z + I_L$$

$$I_Z = I_R - I_L$$

$$I_L = \frac{V_L}{R_L} \quad \text{and} \quad I_R = \frac{V_R}{R} = \frac{V_i - V_L}{R}$$

Application of ZENER Diodes-

- Zener diode is used as a voltage regulator.
- Zener diode is used as a peak clipper in wave shaping circuit.
- Zener diode is used as a fixed reference voltage in transistor biasing circuits.
- Zener diode is used as for meter protection against damage from accidental application of excessive voltage

Q.1 (a) For the Zener diode network of Fig. 46, determine V_L , V_R , I_Z , and P_Z .

(b) Repeat part (a) with $R_L = 3 \text{ k}\Omega$

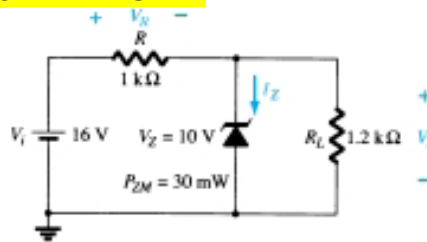
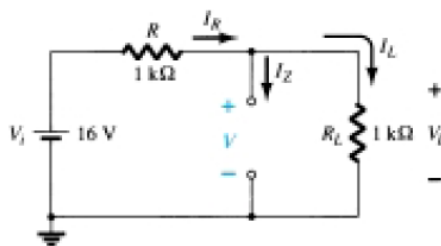


Fig. 46

Answer (a)

$$V = \frac{R_L V_i}{R + R_L} = \frac{1.2 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 8.73 \text{ V}$$



Since $V = 8.73 \text{ V}$ is less than $V_Z = 10 \text{ V}$, the diode is in the “off” state

$$V_L = V = 8.73 \text{ V}$$

$$V_R = V_i - V_L = 16 \text{ V} - 8.73 \text{ V} = 7.27 \text{ V}$$

$$I_Z = 0 \text{ A}$$

$$P_Z = V_Z I_Z = V_Z (0 \text{ A}) = 0 \text{ W}$$

(b)

$$V = \frac{R_L V_i}{R + R_L} = \frac{3 \text{ k}\Omega (16 \text{ V})}{1 \text{ k}\Omega + 3 \text{ k}\Omega} = 12 \text{ V}$$

Since $V = 12 \text{ V}$ is greater than $V_Z = 10 \text{ V}$, the diode is in the “on” state

$$V_L = V_Z = 10 \text{ V}$$

$$V_R = V_i - V_L = 16 \text{ V} - 10 \text{ V} = 6 \text{ V}$$

$$I_L = \frac{V_L}{R_L} = \frac{10 \text{ V}}{3 \text{ k}\Omega} = 3.33 \text{ mA}$$

$$I_R = \frac{V_R}{R} = \frac{6 \text{ V}}{1 \text{ k}\Omega} = 6 \text{ mA}$$

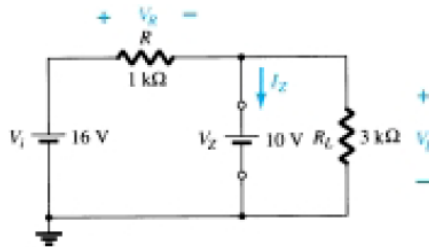
$$I_Z = I_R - I_L \text{ [Eq. (2.18)]}$$

$$= 6 \text{ mA} - 3.33 \text{ mA}$$

$$= 2.67 \text{ mA}$$

The power dissipated,

$$P_Z = V_Z I_Z = (10 \text{ V})(2.67 \text{ mA}) = 26.7 \text{ mW}$$



2.2 Tunnel Diodes

- The tunnel diode is a highly conductive, heavily doped PN-junction diode in which the current induces because of the tunnelling. The tunnelling is the phenomenon of conduction in the semiconductor material in which the charge carrier punches the barrier instead of climbing through it.
- A tunnel diode was discovered by Leo Esaki in the year 1958. He noticed that if a semiconductor diode is doped heavily through impurities, then it will generate negative resistance that means the flow of current across the diode will decreases once the voltage enhances. This is referred to as “Tunneling”.

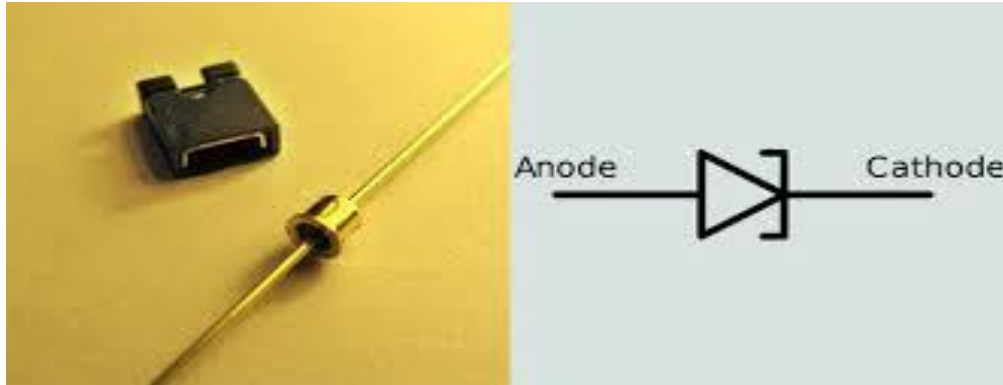


Fig. 47 Tunnel Diode and its symbol

Need to Study Tunnel Diodes

Very high frequency applications using the tunnel diode are possible because the tunneling action occurs so rapidly that there is no transit time effect and therefore no signal distortion. Tunnel diodes are also used extensively in high-speed switching circuits because of the speed of the tunneling action.

Constructional Details of Tunnel Diodes

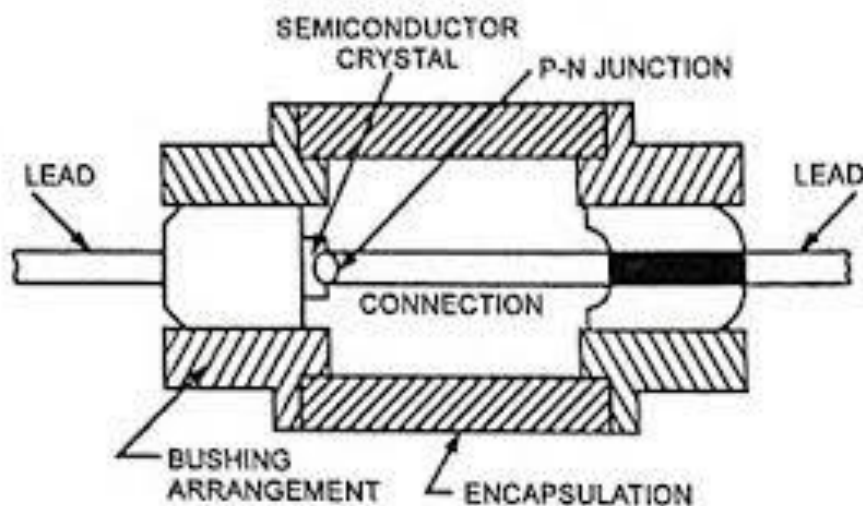


Fig. 48 Construction of Tunnel Diode

- Tunnel Diode are usually fabricated from germanium, gallium arsenide or gallium antimonide.
- Silicon is not used in the tunnel diode construction, because the ratio of peak value of forward current to the value of valley current (i.e., I_P/I_V) is maximum in case of gallium arsenide (approximately 20) and comparatively smaller for germanium (roughly 10) but very small in case of silicon (about 3).
- The source materials are highly doped semiconductor crystals with an impurity concentration of the order of 10^{25} per cubic metre or more.

VI Characteristics of Tunnel Diodes

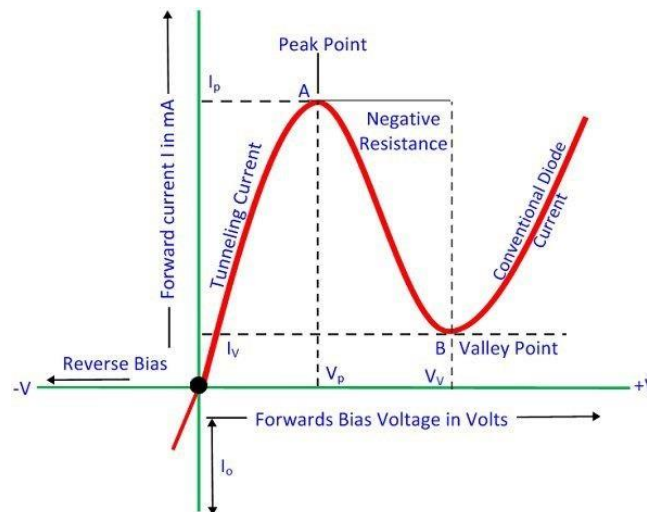


Fig. 49. VI Characteristics of Tunnel Diode

- Due to forward biasing, because of heavy doping conduction happens in the diode. The maximum current that a diode reach is I_p and voltage applied is V_p .
- The current value decreases, when more amount of voltage is applied.
- Current keeps decreasing until it reaches a minimal value.
- The small minimal value of current is I_v . From the above graph, it is seen that from point A to B current reduces when voltage increases. That is the negative resistance region of diode. In this region, tunnel diode produces power instead of absorbing it.

Advantages & Disadvantages of Tunnel Diodes

Advantages

- Long life
- High-speed operation
- Low noise
- Low power consumption

Disadvantages

- Tunnel diodes cannot be fabricated in large numbers
- Being a two-terminal device, the input and output are not isolated from one another.

Applications of Tunnel Diodes

- Tunnel diodes are used as logic memory storage devices.
- Tunnel diodes are used in relaxation oscillator circuits.
- Tunnel diode is used as an ultra-high-speed switch.
- Tunnel diodes are used in FM receivers.

2.2 Schottky Diode

Introduction to Schottky Diode

The Schottky diode is a metal-semiconductor junction diode. A Schottky diode is also known as the hot carrier diode or the Schottky barrier diode. It is formed by the junction of a semiconductor diode with a metal. It has a low forward voltage drop, almost equal to 0.15 to 0.45V. A Schottky diode, like any other diode, conducts current in the forward direction when enough forward voltage is applied.

In a Schottky diode, metals such as aluminium and platinum are used in place of p-type conductors. The Schottky diode was invented by Walter H. Schottky in 1914.

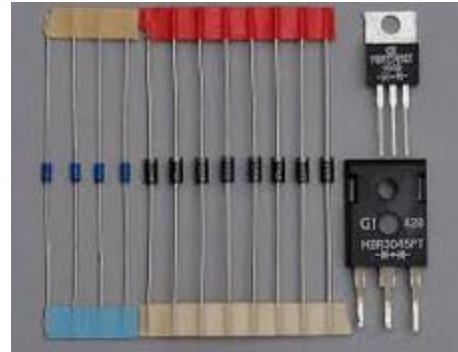
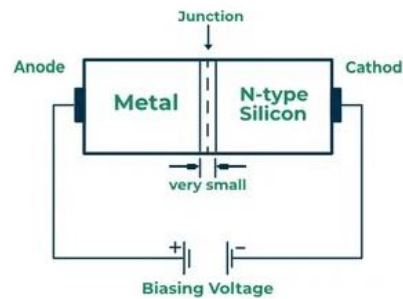


Fig. 50 Schottky Diode



Fig. 51 Schottky Diode Symbol

Characteristics of Schottky Diode

- The Schottky diode is a metal-semiconductor junction diode.
- It has a low forward voltage drop, almost equal to 0.15 to 0.45V.
- These diodes have a high current density.
- In a Schottky diode there is a Schottky barrier, which is a depletion layer formed at the junction of the metal and N-type semiconductor.

Need to Study Schottky Diode

These diodes have a high current density and a low forward voltage drop. This means that in a Schottky diode, less power is wasted than in a PN junction diode. This property of Schottky diodes makes them more suitable for power rectifiers. In this diode, there is a Schottky barrier, which is a depletion layer formed at the

junction of the metal and N-type semiconductor. The electrons have to overcome the potential energy of this Schottky barrier to flow across the diode. The rectified metal-semiconductor junction forms a rectifying Schottky barrier, which is used to make Schottky diodes.

Constructional Details of Schottky Diode

Unlike a normal P-N junction diode, a Schottky diode is equipped with a metallic contact. On further analysis the construction of the diode is like:

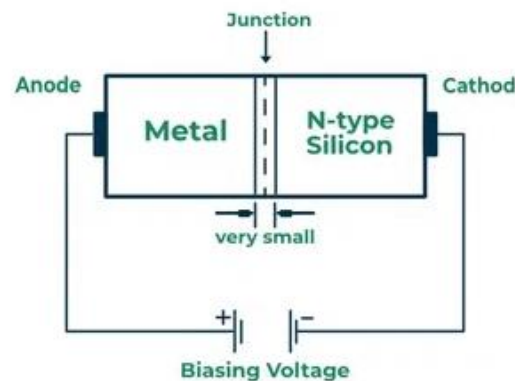


Fig. 52 Schottky Diode Construction

- Schottky diodes are constructed using semiconductor materials. Silicon (Si) is a common choice, although other materials can also be used.
- The differentiating feature in Schottky diode is the metal-to-semiconductor junction. A metal contact (made of metals like Pt, W, Al) is directly connected to the semiconductor material (usually n-type silicon).
- The region beneath the metal contact is n-type semiconductor. It has a higher concentration of electrons (i.e, electrons are the majority charge carriers).
- Schottky diodes have a depletion region near the junction. Since we do not have a p-type region, this depletion region is much thinner than a normal diode.
- The interaction of electrons from the n-type semiconductor with metal atoms is the major driving factor for formation of this region.

Working Principle of Schottky Diode

- A Schottky diode is formed by joining a metal (commonly aluminum or platinum) with a semiconductor material (usually silicon or gallium arsenide) to create a metal-semiconductor junction.
- When the metal comes in contact with the semiconductor, a potential barrier, known as the Schottky barrier, is created at the junction due to the difference in work functions between the metal and semiconductor. This barrier prevents the flow of electrons from the metal into the semiconductor.
- When a forward bias voltage is applied across the Schottky diode (anode connected to the metal side, cathode connected to the semiconductor side), the applied voltage reduces the height of the Schottky barrier.
- The lowered barrier allows electrons to easily flow from the metal into the semiconductor, resulting in a low forward voltage drop (typically 0.2 to 0.4 volts for silicon Schottky diodes). This characteristic minimizes power dissipation and makes the diode suitable for low-voltage applications.
- Schottky diodes exhibit very fast switching action due to the absence of a depletion region, which is present in traditional PN junction diodes. This makes them ideal for high-frequency applications where rapid switching is required.
- In the reverse bias condition (positive voltage on the semiconductor side, negative voltage on the metal side), the Schottky diode exhibits a small reverse leakage current due to thermionic emission of electrons over the lowered Schottky barrier.
- Unlike PN junction diodes, Schottky diodes have virtually no reverse recovery time. This means there is no delay when transitioning from the conducting to non-conducting state, making them suitable for high-frequency rectification.

V-I Characteristics of Schottky Diode

- The V-I characteristics of Schottky diodes are very much similar to the PN junction diode. Current is the dependent variable while voltage is the independent variable in the Schottky diode.

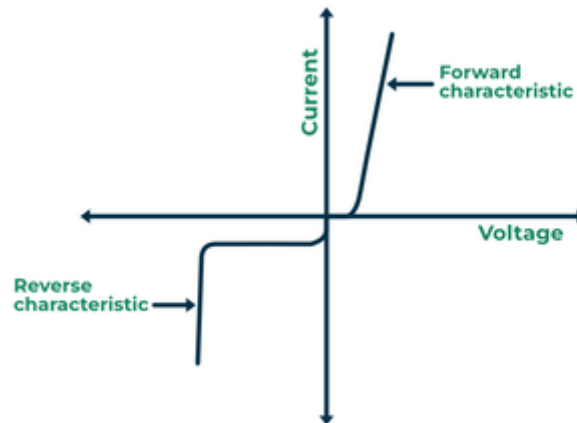


Fig. 53 V-I Characteristic of Schottky Diode

- Forward Bias (F): When a forward bias voltage is applied across a Schottky diode (positive voltage on the anode, negative voltage on the cathode), the V-I characteristics show that it conducts current very quickly with a relatively low forward voltage drop. This rapid conduction is due to the low Schottky barrier at the metal-semiconductor junction. The forward current increases exponentially with voltage, similar to a standard diode.
- Reverse Bias (R): In the reverse bias condition (positive voltage on the cathode, negative voltage on the anode), the V-I characteristics of a Schottky diode display a small reverse current, often referred to as the leakage current. This reverse current is typically larger than that of a silicon PN junction diode because of the absence of the depletion region found in PN junctions. However, it is still relatively low compared to other diode types.
- Breakdown Voltage: Schottky diodes do not exhibit a well-defined breakdown voltage like some other diode types. Instead, they have a reverse breakdown mechanism known as "avalanche breakdown." When the reverse voltage exceeds

a certain level, the diode can experience avalanche breakdown, leading to a sudden increase in reverse current.

- **Cut-in Voltage:** The cut-in voltage (also called the knee voltage) for a Schottky diode is typically very close to zero volts, which means it starts conducting at very low forward bias voltages.

Advantages & Disadvantages of Schottky Diodes

Advantages

- **Low Turn-On voltage:** The voltage required to turn on a Schottky diode is comparatively low as compared to other diodes.
- **Low Forward Voltage Drop:** Due to this property of diode it finds its application in Power Circuits because it is well adapted due to low voltage drop. This property also makes them best devices to be used in Power Rectifier Applications with high current density.
- **Fast Recovery Time:** Due to the semiconductor contact, Schottky diodes show a very fast recovery time making the circuits very fast. Thereby these diodes are used in Fast-Clamp Diode Applications where base junction is forward biased.
- **Low Loss Energy:** Schottky diode shows a very low energy loss even at higher frequencies. Due to this Schottky diodes are generally used in Solar Cells. Solar cells generally store energy and Schottky diodes prevent them from discharging.

Disadvantages

- **High Reverse Leakage Current:** A significant drawback is their higher reverse leakage current compared to traditional diodes, which can increase with temperature.
- **Low Reverse Breakdown Voltage:** Schottky diodes typically have a lower maximum reverse bias voltage they can withstand before breaking down, limiting their use in high-voltage applications.

- **Temperature Sensitivity:** The reverse leakage current shows a positive temperature characteristic, meaning it increases with rising temperature, which can lead to thermal breakdown.
- **Limited Application in Some Areas:** Due to the leakage current and voltage limitations, they are less suitable for applications requiring precise voltage regulation, such as peak detection circuits or reverse voltage polarity protection.

Applications of Schottky Diodes

Schottky diodes are very useful in the electronics industry. Undermentioned are some major areas of their use:

- Solar cells
- Power OR electric circuits
- Power rectifier (like bridge rectifier)
- RF mixer and detector diode

2.4 Varactor Diodes

Introduction to Varactor Diodes

- The varactor diode is also known as variable capacitance diode, variable reactance diode or tuning diode.
- The varactor diode always function in reverse biasing condition.
- Its doping level is high to increase capacitance of its depletion region, here the capacitance term is used because its operation is similar to the capacitor.



Fig. 54. Varactor Diodes

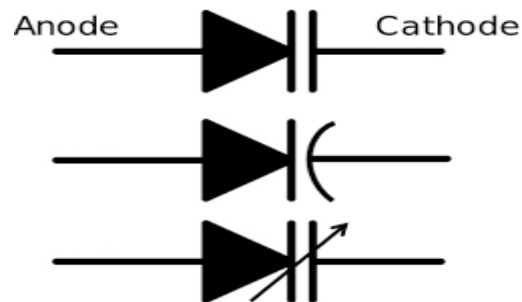


Fig. 55 Symbol of Varactor Diodes

Need to Study Varactor Diodes

- Varactor or varicap diodes are used mainly in radio frequency or RF circuit designs to provide voltage controlled variable capacitance.
- These electronic components can be used in a whole variety of ways where a capacitance level needs to be controlled by a voltage, and without them, many of the capabilities we enjoy in radios, mobile phones and many other devices would not be possible.

Working Principle of Varactor Diodes

- It is formed of P-type and N-type semiconductor and reverse biasing is applied to it.

- Varactor diode produces capacitance effect, this is because its cathode and anode terminals act as the plates of the capacitor and the region between them acts as the dielectric medium.
- The majority carriers in an N-type semiconductor are electrons and the majority carriers in a P-type semiconductor are holes. At the junction, the electrons and holes recombine.
- Due to which immobile ions accumulate at the junction. And no more current can flow due to majority carriers. Thus, the depletion region is formed.
- The depletion region is called so because it is depleted of charge carriers i.e. the majority carriers are absent in the depletion region. This works as a dielectric layer and P and N-type semiconductor work as plates of a capacitor.
- When the reverse bias is applied to a P-N junction, the width of the depletion layer increases. And with the increase of reverse voltage, gradually the depletion layer increases even more. Thus, the depletion region creates Transition capacitance C_T .

$$C_T = \epsilon A / W$$

- Here, C_T is Transition capacitance, ϵ is dielectric constant, A is the area of plates of the capacitor and W is the width of the depletion layer.

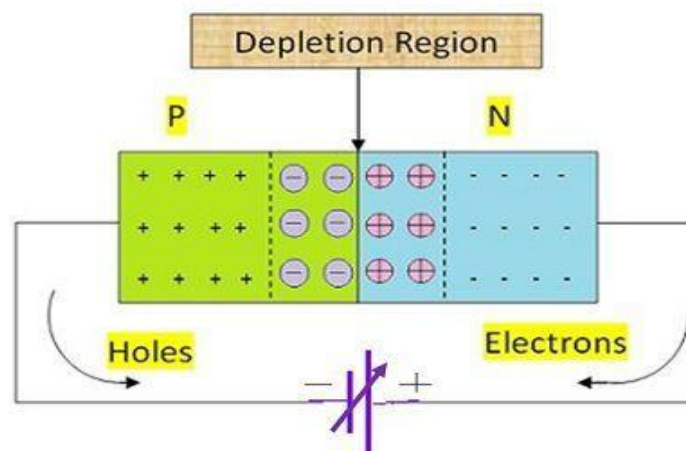


Fig. 56 Working Principle of Varactor Diodes

- It is evident from the above relation that transition capacitance is inversely proportional to the width of the depletion layer. Thus, if we want the high magnitude

of capacitance the width should be small. And the width will be small if we will apply low reverse voltage.

- Similarly, if we require low capacitance the width should be large and to increase the width the reverse voltage applied should be high. Thus, this width can be controlled with applied reverse voltage.
- The tuning range of the capacitor varies with the doping level of the diode. For abrupt doping junction the doping will be uniform, but for the hyper abrupt junction, the doping profile will be non-uniform.

Transition capacitance can be given as (Varactor diode Capacitance)

$$C_T = \frac{K}{(V_R + V_K)^n}$$

where,

V_K = knee voltage

V_R = applied reverse voltage

K = constant depending on semiconductor material

n = constant called the gradient coefficient.

V-I Characteristics of Varactor Diodes

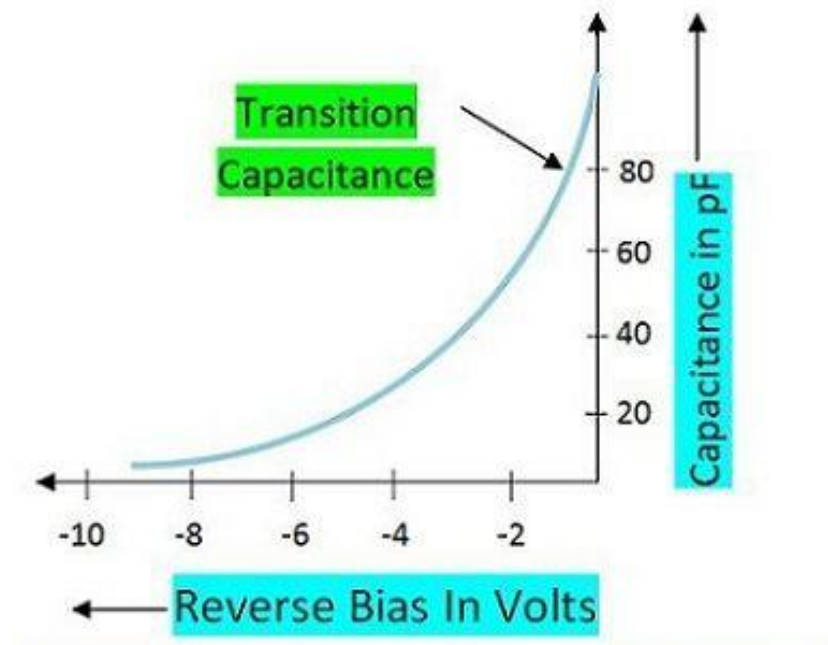


Fig..57. Characteristics of Varactor Diodes

Advantages & Disadvantages of Varactor Diodes

Advantages

- **Low Noise:** It generates less noise as compared to the other P-N junction diode. Thus, the power loss due to noise is low in varactor diodes.
- **Portability:** It is portable due to the small size and lightweight.
- **Reliability:** It is more reliable than other P-N junction diodes.
- **Economical:** It is a low-cost diode thus, it is economical to use in various applications.

Disadvantages

- These are specially designed to work in the reverse biased mode, it possesses the least significance when operated in forward biasing.

Applications of Varactor Diodes

- **Television receivers:** Varactor diodes are used as tuned capacitors and have replaced mechanically tuned capacitors in various applications. It is used in television in the resonant tank circuit.
- **Radio receivers:** Radio receivers also use this diode for tuning purposes.
- **Frequency Multiplier:** It is also used as a frequency multiplier in various electronic circuits.
- **Phase Locked Loops:** It is used in Phase locked loop for frequency modulation. Varactor diodes help in achieving frequency modulation. Thus, in communication devices varactor diodes are significant.
- **Voltage controlled oscillators:** Voltage control oscillators are used extensively in transmission and receiving circuits in communication. And varactor diode plays a significant role in construction of voltage-controlled oscillator.
- **Parametric Amplifiers:** It is used in parametric amplifier as a significant component.

2.5 Photodiodes

Introduction to Photodiodes

The purpose of a photodiode is to generate current that is proportional to the intensity of visible, infrared, or ultraviolet light. The technical term for light intensity as measured by a photodiode is illuminance.

A photodiode has transparent packaging that allows light to reach the pn junction, and in a properly designed photodiode circuit, incident light will create precise variations in the amount of current flowing through the photodiode.



Fig. 3.4.1. Photodiodes

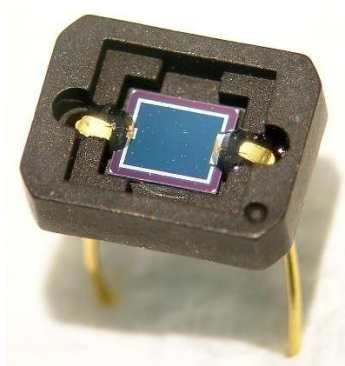


Fig. 3.4.2. Close up of a Photodiode

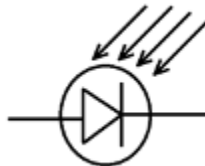


Fig. 3.4.3. Symbol of Photodiode

Need to Study Photodiodes

Photodiodes are frequently used photodetectors, which have largely replaced the formerly used vacuum phototubes. They are semiconductor devices which contain a p-n junction, and often an intrinsic (undoped) layer between n and p layers. Devices with an intrinsic layer are called p-i-n or PIN photodiodes. Light absorbed in the depletion region or the intrinsic region generates electron-hole pairs, most

of which contribute to a photocurrent. The photocurrent can be quite precisely proportional to the absorbed (or incident) light intensity over a wide range of optical powers.

Constructional Details of Photodiodes

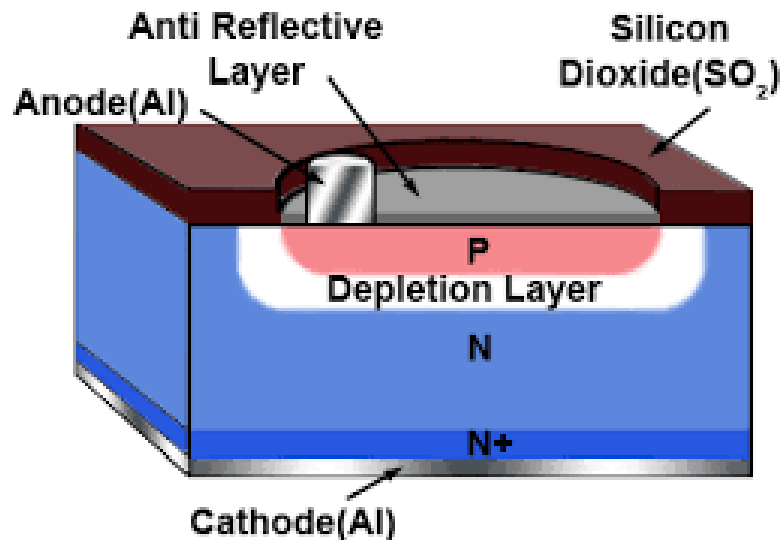


Fig. 3.4.4. Construction of Photodiode

- The surface of a layer of N type is bombarded with P type silicon ions to produce a P type layer about 1 μm thick.
- During the formation of the diode, electrons from the N type layer are attracted into the P type material and holes from the P type are attracted into the N type layer, resulting in the removal of free charge carriers close to the PN junction, so creating a depletion layer.
- The (light facing) top of the diode is protected by a layer of Silicon Dioxide (SiO_2) in which there is a window for light to shine on the semiconductor.
- This window is coated with a thin anti-reflective layer of Silicon Nitride (SiN) to allow maximum absorption of light and an anode connection of aluminium (Al) is provided to the P type layer.

- Beneath the N type layer is a more heavily doped N+ layer to provide a low resistance connection to the cathode.

Working Principle of Photodiodes

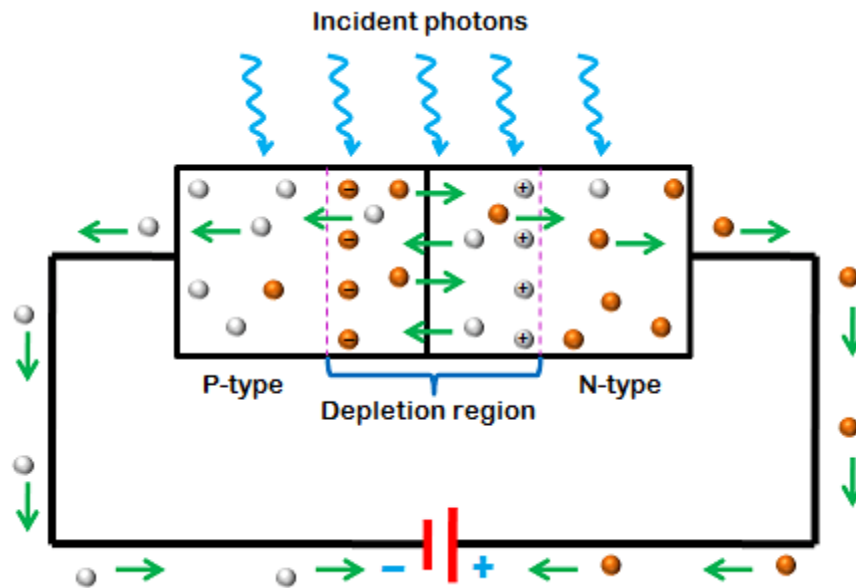


Fig. 3.4.5. Working Principle of Photodiode

- The working principle of a photodiode is, when a photon of ample energy strikes the diode, it makes a couple of an electron-hole. This mechanism is also called the **inner photoelectric effect**.
- If the absorption arises in the depletion region junction, then the carriers are removed from the junction by the inbuilt electric field of the depletion region.
- Therefore, holes in the region move toward the anode, and electrons move toward the cathode, and a photocurrent will be generated. The entire current through the diode is the sum of the absence of light and the photocurrent. So, the absent current must be reduced to maximize the sensitivity of the device.

V-I Characteristics of Photodiodes

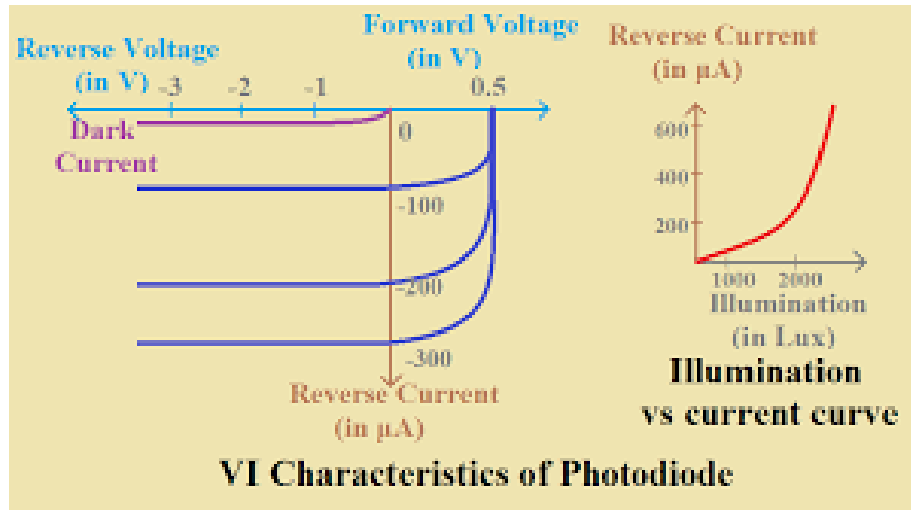


Fig. 3.4.6. Working Principle of Photodiode

The value of reverse current is independent of reverse voltage. It depends on the intensity of the illumination of light. The variation of current w.r.t variation of illumination is shown above. The reverse current is almost zero when the illumination of light is zero. This minimum reverse current is known as Dark Current.

Types of Photodiodes

Although there are numerous types of photodiode available in the market and they all work on the same basic principles, though some are improved by other effects. The working of different types of photodiodes works in a slightly different way, but the basic operation of these diodes remains the same. The types of photodiodes can be classified based on their construction and functions as follows.

- PN Photodiode
- Schottky Photo Diode
- PIN Photodiode
- Avalanche Photodiode

Advantages & Disadvantages of Photodiodes

Advantages

- Less resistance
- Quick and high operation speed
- Long life span.
- Fastest photodetector.
- Spectral response is good.
- Doesn't use high voltage.
- Frequency response is good.
- Solid and low-weight.
- It is extremely responsive to the light.
- Dark current is less.
- High quantum efficiency.
- Less noise.

Disadvantages

- Temperature stability is poor.
- Change within current is extremely little, therefore may not be enough to drive the circuit.
- The active area is small.
- Usual PN junction photodiode includes a high response time

Applications of Photodiodes

- Are used as switching devices in switching circuits.
- Light detectors, light-operated switches.
- Also used in electronic control circuits like counting devices and communication systems.

- Are used as encoders, demodulators.
-
- Are used in computer card punching and tapes